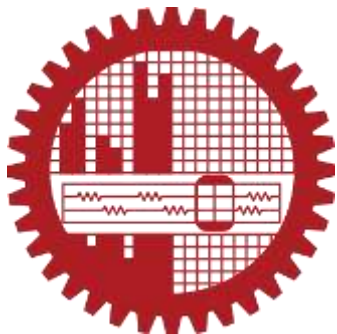
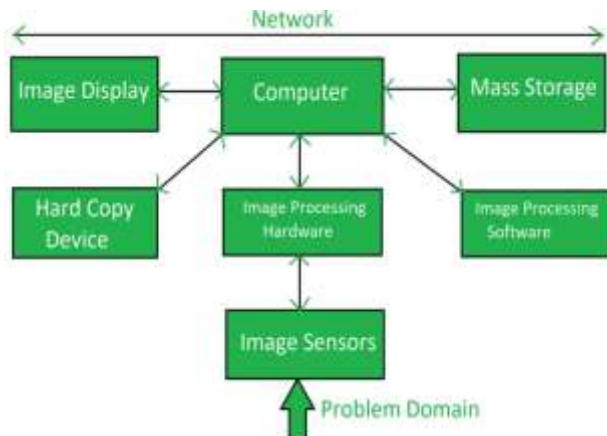
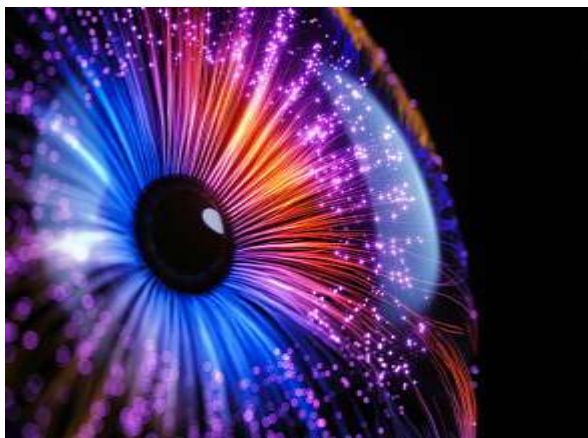


Digital Image Processing



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Topic 4

Image Filtering in Frequency Domain

Introduction

Image filtering modifies an image's pixel values using mathematical operations, often with a small matrix called a **kernel**, to enhance quality, reduce noise, sharpen details, or extract features like edges, achieved through **spatial domain** (direct pixel manipulation) or **frequency domain** (transforming to frequency, filtering, transforming back) methods, with common examples being **low-pass filters** (smoothing) and **high-pass filters** (sharpening). It is a fundamental technique used to enhance or modify an image by altering its pixel values through mathematical operations. It is primarily used for noise reduction, smoothing, edge detection, and sharpening.

- **Kernel/Mask:** A small matrix (e.g., 3x3) that moves across the image, pixel by pixel.
- **Convolution:** For each pixel, the kernel's values are multiplied by the corresponding neighboring pixel values, and the results are summed to create a new pixel value.

Key Goals

- Improve visual quality.
- Extract meaningful information.
- Prepare images for further analysis (e.g., machine learning).

Spatial Domain Filtering

This method involves direct manipulation of image pixels using a **kernel** (a small matrix) that slides across the image.

- ❖ **Linear Filters:** Perform weighted sums of neighboring pixels.
 - **Mean/Averaging Filter:** Smooths images by replacing a pixel with the average of its neighbors, effective for general noise but blurs edges.
 - **Gaussian Filter:** A weighted average that gives more importance to central pixels, resulting in smoother blurring with better edge preservation than mean filters.
 - **Sharpening Filters:** Use first or second derivatives (e.g., Laplacian, Sobel, Prewitt) to highlight edges and fine details.
- ❖ **Non-Linear Filters:** Based on more complex rules rather than simple weighted sums.
 - **Median Filter:** Replaces the center pixel with the median value of the neighborhood. It is exceptionally effective at removing "salt-and-pepper" noise while preserving edges.
 - **Min/Max Filters:** Used to find the darkest or brightest points in a neighborhood, often for morphological analysis.

Spatial Domain Filters

Smoothing Filters (Low-Pass): Reduce noise and blur details by averaging pixel values.

Mean/Average Filter: Replaces pixel with the average of its neighbors (e.g., moving average).

Gaussian Filter: Uses a Gaussian function for smoother blurring, effective for random noise.

Order-Statistic Filters (Non-linear): Effective for impulse (salt-and-pepper) noise.

Median Filter: Replaces pixel with the median value of its neighbors, preserving edges better than mean filters.

Sharpening Filters (High-Pass): Enhance edges and details (e.g., Laplacian filter).

Frequency Domain Filtering

This approach transforms the image using the **Fourier Transform**, modifies its frequency components, and then converts it back.

- **Low-Pass Filters (LPF):** Attenuate high-frequency components to achieve smoothing and noise reduction.
- **High-Pass Filters (HPF):** Attenuate low-frequency components to enhance edges and sharpen details.
- **Band-Pass/Band-Reject Filters:** Isolate or remove specific frequency ranges to target specific patterns or interference.

Advanced Filtering Trends

- **Deep Learning-Based Filters:** Convolutional Neural Networks (CNNs) and Generative Adversarial Networks (GANs) are now widely used for high-accuracy denoising, super-resolution, and complex image restoration.
- **Edge-Preserving Filters:** Advanced algorithms like **Bilateral Filters** or **Joint Image Filtering** use additional data (like depth maps) to smooth images without blurring critical boundaries.
- **Real-Time Mobile Filtering:** Modern techniques prioritize efficiency for real-time applications in Augmented Reality (AR) and mobile photography.

Common Applications

- **Noise Reduction:** Removing unwanted artifacts (e.g., from sensors).
- **Image Enhancement:** Blurring for artistic effects or privacy (pixelation), sharpening details, adjusting contrast.
- **Feature Extraction:** Detecting edges, corners, or textures for computer vision tasks.

Main Topics

- Fourier Transform
- Filtering in Fourier Transform Domain

Fourier Series and Fourier Transform

- Fourier Series

Any periodic function can be expressed as the sum of sines and /or cosines of different frequencies, each multiplied by a different coefficients

- Fourier Transform

Any function that is not periodic can be expressed as the integral of sines and /or cosines multiplied by a weighing function

Fourier Series

A function $f(t)$ of a continuous variable t that is periodic with period, T , can be expressed as the sum of sines and cosines multiplied by appropriate coefficients

$$f(t) = \sum_{n=-\infty}^{\infty} c_n e^{j\frac{2\pi n}{T}t}$$

where

$$c_n = \frac{1}{T} \int_{-T/2}^{T/2} f(t) e^{-j\frac{2\pi n}{T}t} dt \quad \text{for } n = 0, \pm 1, \pm 2, \dots$$

Fourier Series: Example

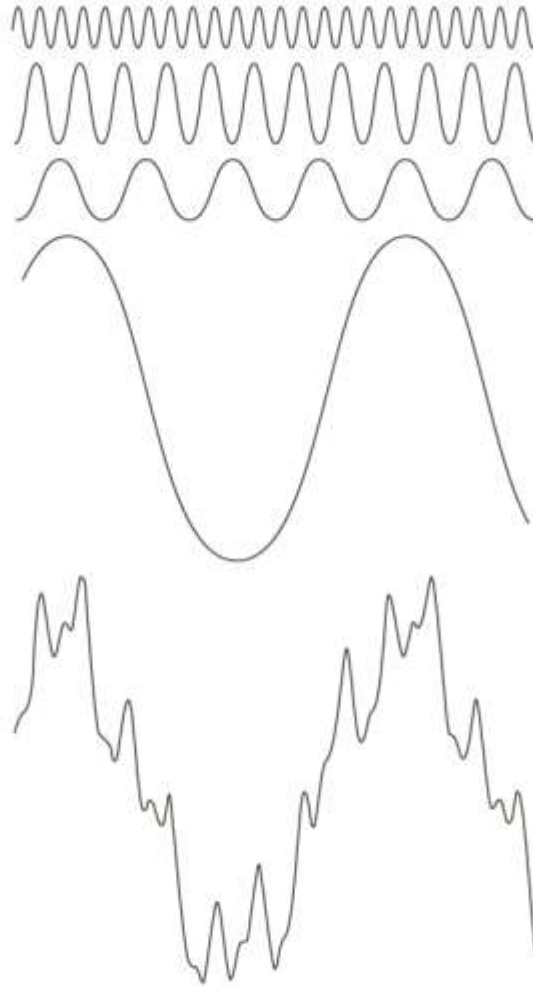


FIGURE 4.1 The function at the bottom is the sum of the four functions above it. Fourier's idea in 1807 that periodic functions could be represented as a weighted sum of sines and cosines was met with skepticism.

Fourier Transform: One Continuous Variable

The *Fourier Transform* of a continuous function $f(t)$

$$F(\mu) = \mathfrak{F}\{f(t)\} = \int_{-\infty}^{\infty} f(t)e^{-j2\pi\mu t} dt$$

The *Inverse Fourier Transform* of $F(\mu)$

$$f(t) = \mathfrak{F}^{-1}\{F(\mu)\} = \int_{-\infty}^{\infty} F(\mu)e^{j2\pi\mu t} d\mu$$

Fourier Transform and Convolution

The convolution of two functions is denoted by the operator $\star$

$$\begin{aligned}f(t) \star h(t) &= \int_{-\infty}^{\infty} f(\tau)h(t-\tau)d\tau \\ \mathfrak{F}\{f(t) \star h(t)\} &= \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} f(\tau)h(t-\tau)d\tau \right] e^{-j2\pi\mu t} dt \\ &= \int_{-\infty}^{\infty} f(\tau) \left[\int_{-\infty}^{\infty} h(t-\tau)e^{-j2\pi\mu t} dt \right] d\tau \\ &= \int_{-\infty}^{\infty} f(\tau) \left[H(\mu)e^{-j2\pi\mu\tau} \right] d\tau \\ &= H(\mu) \int_{-\infty}^{\infty} f(\tau)e^{-j2\pi\mu\tau} d\tau \\ &= H(\mu)F(\mu)\end{aligned}$$

Fourier Transform and Convolution

Fourier Transform Pairs

$$f(t) \star h(t) \Leftrightarrow H(\mu)F(\mu)$$

$$f(t)h(t) \Leftrightarrow H(\mu) \star F(\mu)$$

Discrete Fourier Transform (DFT) of One Variable

$$F(\mu) = \sum_{x=0}^{M-1} f(x) e^{-j2\pi\mu x/M}, \quad \mu = 0, 1, \dots, M-1$$

$$f(x) = \frac{1}{M} \sum_{\mu=0}^{M-1} F(\mu) e^{j2\pi\mu x/M}, \quad x = 0, 1, 2, \dots, M-1$$

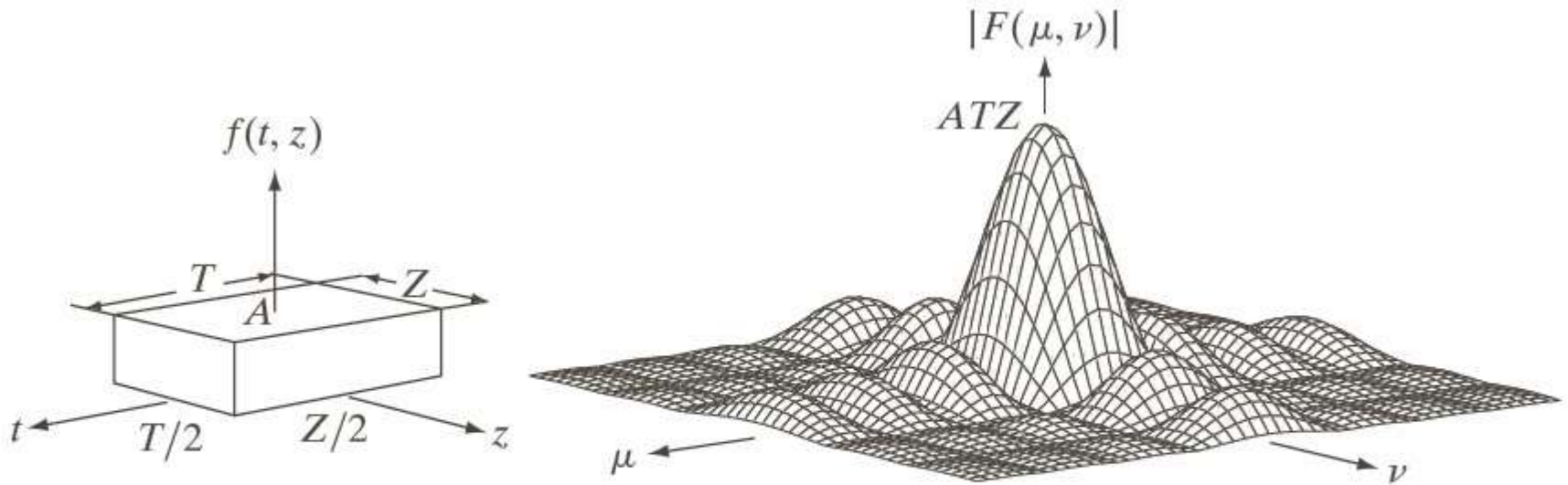
2-D Fourier Transform: Continuous

$$F(\mu, \nu) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(t, z) e^{-j2\pi(\mu t + \nu z)} dt dz$$

and

$$f(t, z) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} F(\mu, \nu) e^{j2\pi(\mu t + \nu z)} d\mu d\nu$$

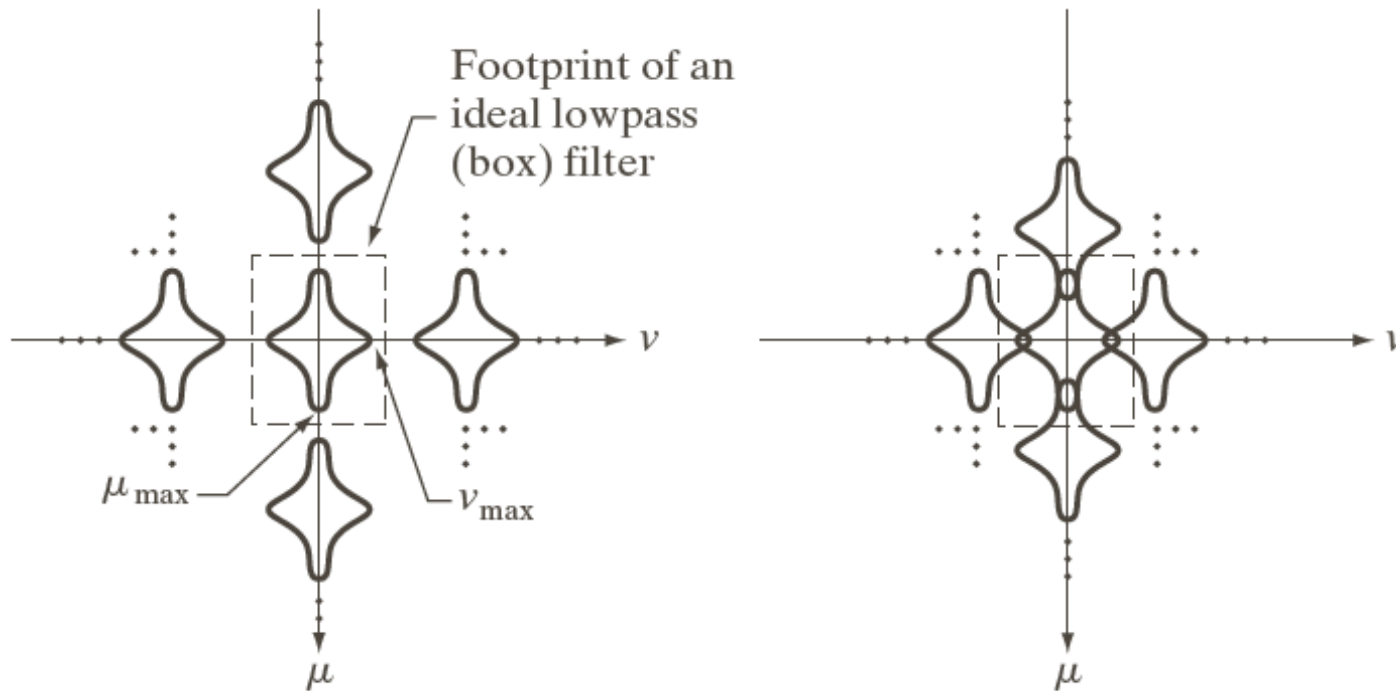
$$\begin{aligned}
 F(\mu, \nu) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(t, z) e^{-j2\pi(\mu t + \nu z)} dt dz \\
 &= \int_{-T/2}^{T/2} \int_{-Z/2}^{Z/2} A e^{-j2\pi(\mu t + \nu z)} dt dz \\
 &= ATZ \left[\frac{\sin(\pi\mu T)}{\pi\mu T} \right] \left[\frac{\sin(\pi\nu T)}{\pi\nu T} \right]
 \end{aligned}$$



a b

FIGURE 4.13 (a) A 2-D function, and (b) a section of its spectrum (not to scale). The block is longer along the t -axis, so the spectrum is more “contracted” along the μ -axis.

2-D Sampling and 2-D Sampling Theorem



a b

FIGURE 4.15
Two-dimensional
Fourier transforms
of (a) an over-
sampled, and
(b) under-sampled
band-limited
function.

Aliasing in Images: Example

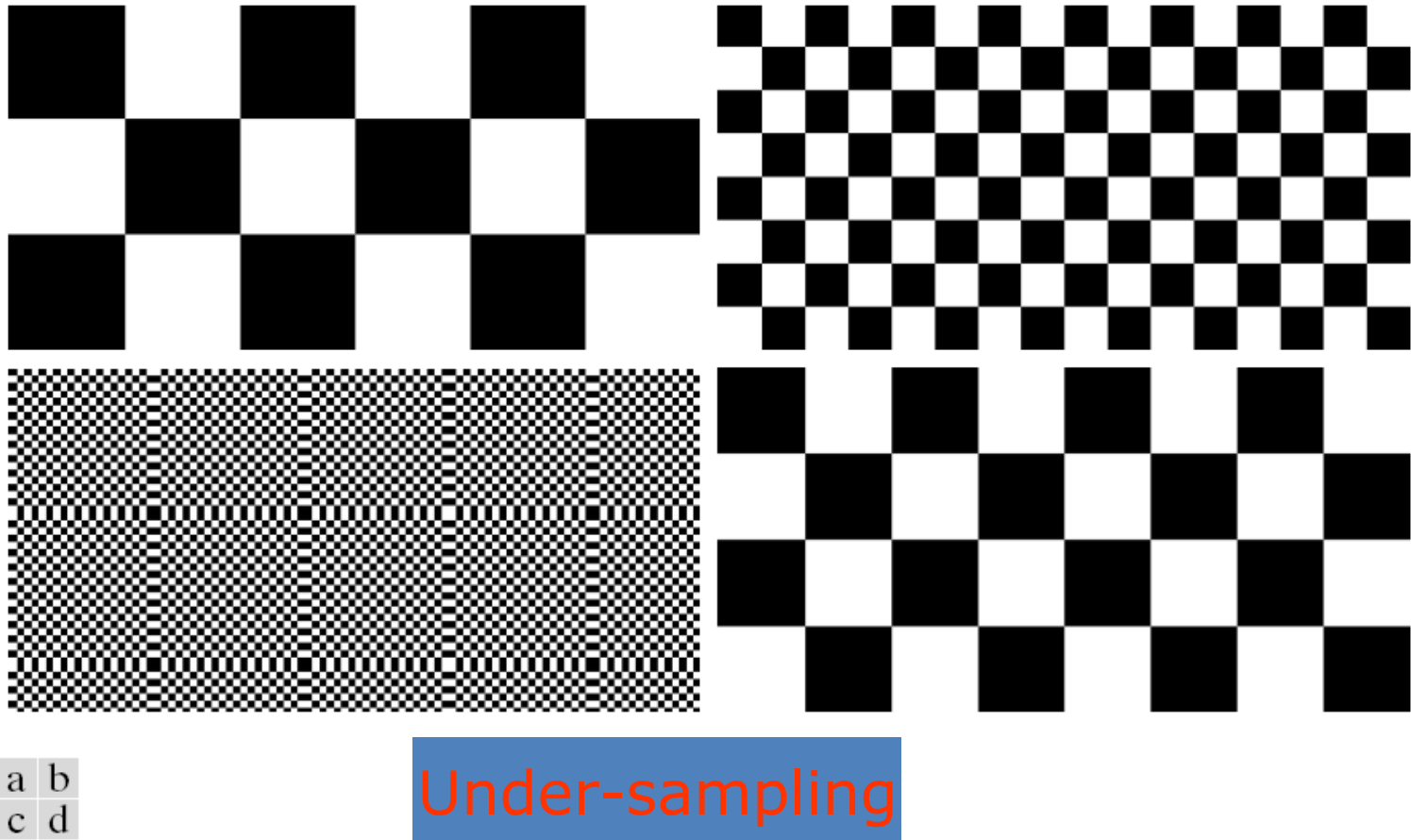


FIGURE 4.16 Aliasing in images. In (a) and (b), the lengths of the sides of the squares are 16 and 6 pixels, respectively, and aliasing is visually negligible. In (c) and (d), the sides of the squares are 0.9174 and 0.4798 pixels, respectively, and the results show significant aliasing. Note that (d) masquerades as a “normal” image.

Aliasing in Images: Example

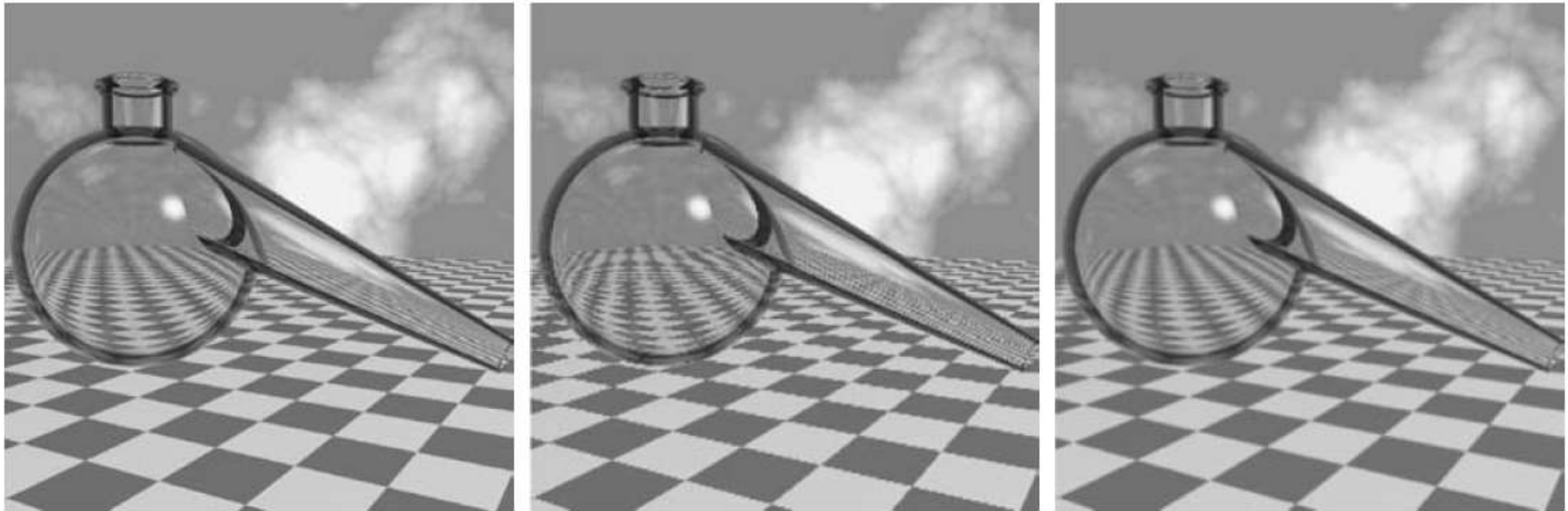


a b c

Re-sampling

FIGURE 4.17 Illustration of aliasing on resampled images. (a) A digital image with negligible visual aliasing. (b) Result of resizing the image to 50% of its original size by pixel deletion. Aliasing is clearly visible. (c) Result of blurring the image in (a) with a 3×3 averaging filter prior to resizing. The image is slightly more blurred than (b), but aliasing is not longer objectionable. (Original image courtesy of the Signal Compression Laboratory, University of California, Santa Barbara.)

Aliasing in Images: Example



a b c

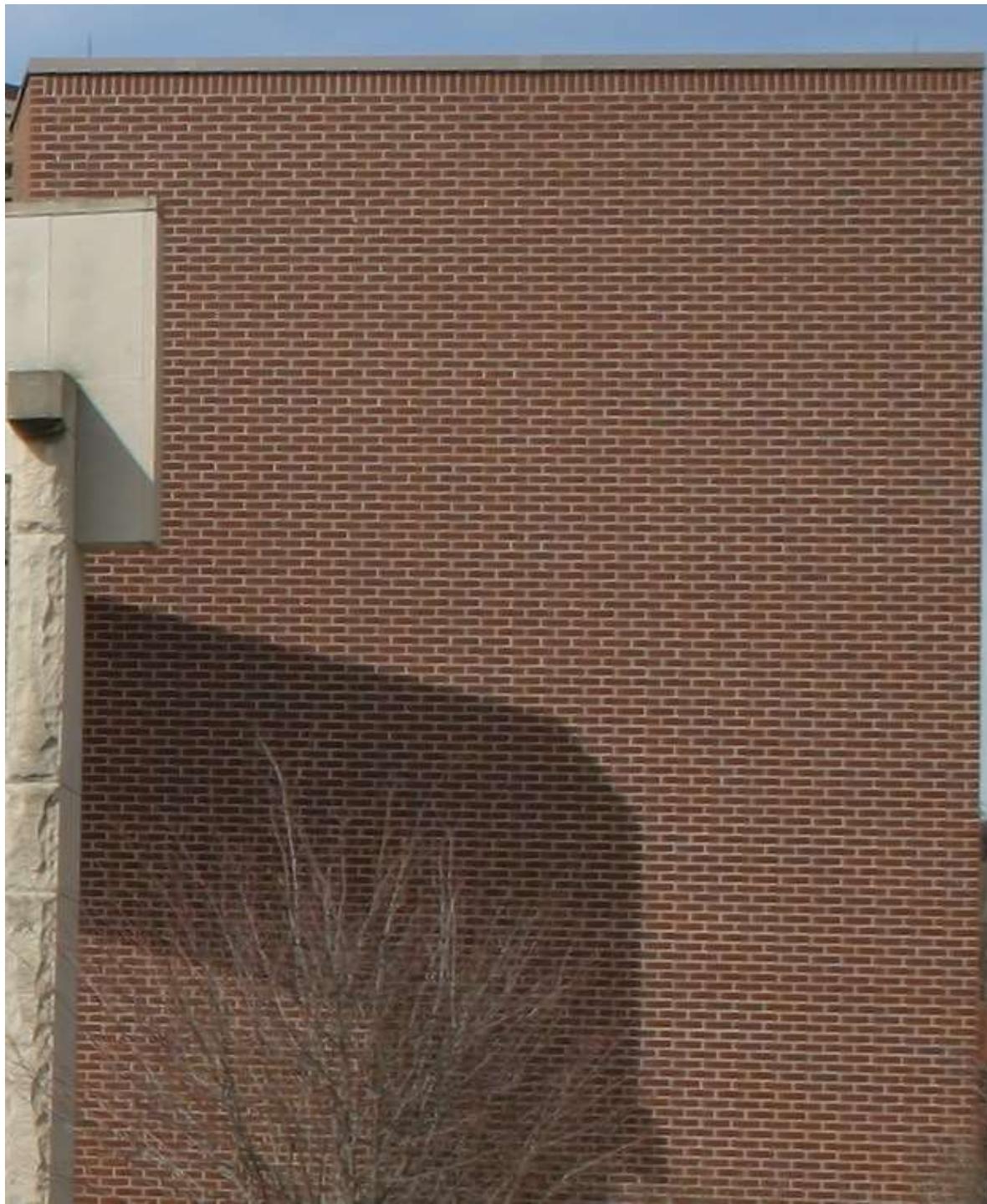
Re-sampling

FIGURE 4.18 Illustration of jaggies. (a) A 1024×1024 digital image of a computer-generated scene with negligible visible aliasing. (b) Result of reducing (a) to 25% of its original size using bilinear interpolation. (c) Result of blurring the image in (a) with a 5×5 averaging filter prior to resizing it to 25% using bilinear interpolation. (Original image courtesy of D. P. Mitchell, Mental Landscape, LLC.)

Moiré patterns

- Moiré patterns are often an undesired artifact of images produced by various digital imaging and computer graphics techniques
 - e. g., when scanning a halftone picture or ray tracing a checkered plane. This cause of moiré is a special case of aliasing, due to under-sampling a fine regular pattern







A moiré pattern
formed by
incorrectly down-
sampling the
former image

2-D Discrete Fourier Transform and Its Inverse

DFT:

$$F(\mu, \nu) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi(\mu x/M + \nu y/N)}$$

$$\mu = 0, 1, 2, \dots, M-1; \nu = 0, 1, 2, \dots, N-1;$$

$f(x, y)$ is a digital image of size $M \times N$.

IDFT:

$$f(x, y) = \frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} F(\mu, \nu) e^{j2\pi(\mu x/M + \nu y/N)}$$

Properties of the 2-D DFT

relationships between spatial and frequency intervals

Let ΔT and ΔZ denote the separations between samples, then the separations between the corresponding discrete, frequency domain variables are given by

$$\Delta\mu = \frac{1}{M\Delta T}$$

and
$$\Delta\nu = \frac{1}{N\Delta Z}$$

Properties of the 2-D DFT

translation and rotation

$$f(x, y)e^{j2\pi(\mu_0 x/M + \nu_0 y/N)} \Leftrightarrow F(\mu - \mu_0, \nu - \nu_0)$$

and

$$f(x - x_0, y - y_0) \Leftrightarrow F(\mu, \nu)e^{-j2\pi(\mu x_0/M + \nu y_0/N)}$$

Using the polar coordinates

$$x = r \cos \theta \quad y = r \sin \theta \quad \mu = \omega \cos \varphi \quad \nu = \omega \sin \varphi$$

results in the following transform pair:

$$f(r, \theta + \theta_0) \Leftrightarrow F(\omega, \varphi + \theta_0)$$

Properties of the 2-D DFT

periodicity

2-D Fourier transform and its inverse are infinitely periodic

$$F(\mu, \nu) = F(\mu + k_1 M, \nu) = F(\mu, \nu + k_2 N) = F(\mu + k_1 M, \nu + k_2 N)$$

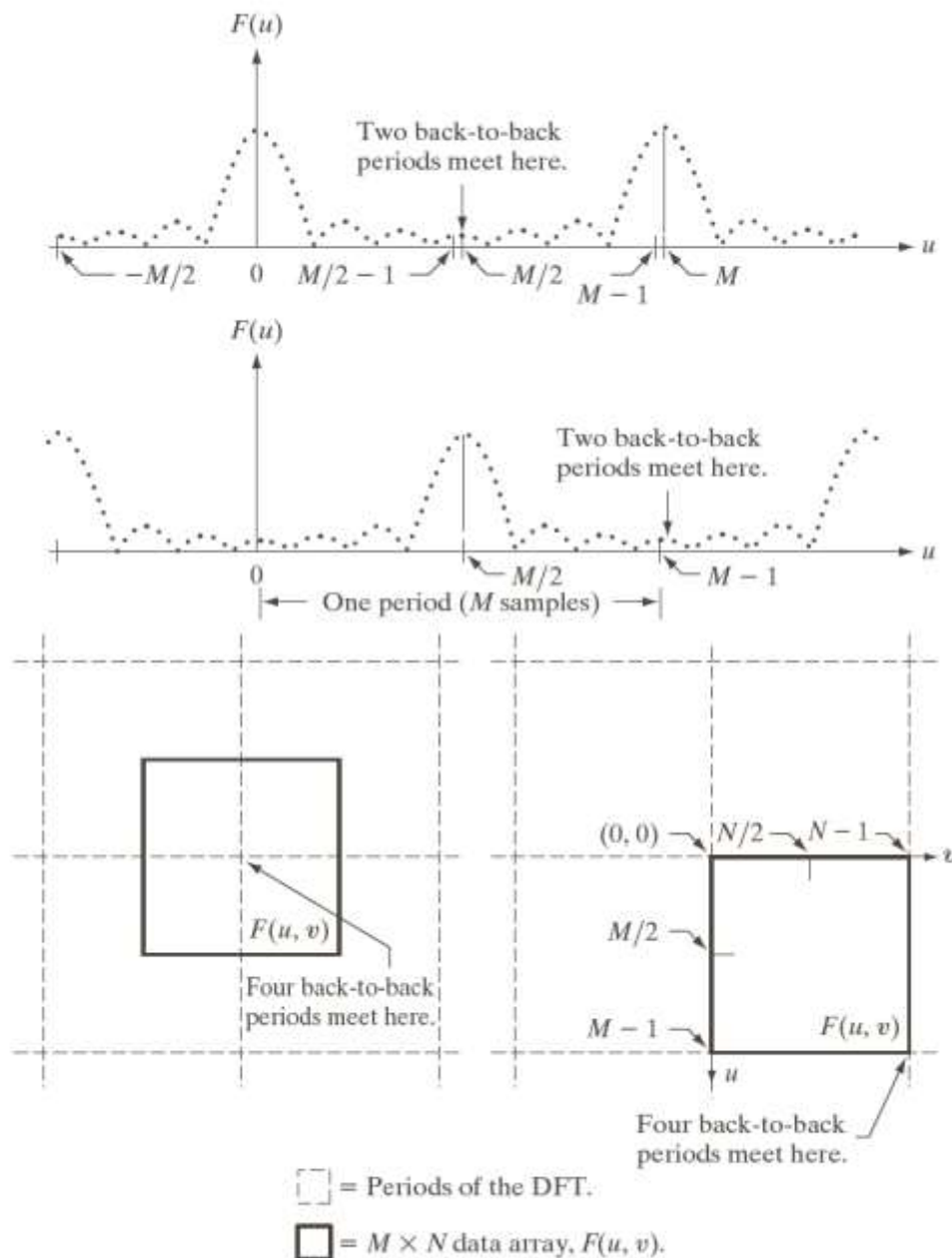
$$f(x, y) = f(x + k_1 M, y) = f(x, y + k_2 N) = f(x + k_1 M, y + k_2 N)$$

$$f(x) e^{j2\pi(\mu_0 x / M)} \Leftrightarrow F(\mu - \mu_0)$$

$$\mu_0 = M / 2, \quad f(x)(-1)^x \Leftrightarrow F(\mu - M / 2)$$

$$f(x, y)(-1)^{x+y} \Leftrightarrow F(\mu - M / 2, \nu - N / 2)$$

Periodicity



a
b
c d

FIGURE 4.23

Centering the Fourier transform.

(a) A 1-D DFT showing an infinite number of periods.

(b) Shifted DFT obtained by multiplying $f(x)$ by $(-1)^x$ before computing $F(u)$.

(c) A 2-D DFT showing an infinite number of periods.

The solid area is the $M \times N$ data array, $F(u, v)$, obtained with Eq. (4.5-15). This array consists of four quarter periods.

(d) A Shifted DFT obtained by multiplying $f(x, y)$ by $(-1)^{x+y}$ before computing $F(u, v)$. The data now contains one complete, centered period, as in (b).

Symmetry

Spatial Domain [†]		Frequency Domain [†]
1)	$f(x, y)$ real	$\Leftrightarrow F^*(u, v) = F(-u, -v)$
2)	$f(x, y)$ imaginary	$\Leftrightarrow F^*(-u, -v) = -F(u, v)$
3)	$f(x, y)$ real	$\Leftrightarrow R(u, v)$ even; $I(u, v)$ odd
4)	$f(x, y)$ imaginary	$\Leftrightarrow R(u, v)$ odd; $I(u, v)$ even
5)	$f(-x, -y)$ real	$\Leftrightarrow F^*(u, v)$ complex
6)	$f(-x, -y)$ complex	$\Leftrightarrow F(-u, -v)$ complex
7)	$f^*(x, y)$ complex	$\Leftrightarrow F^*(-u - v)$ complex
8)	$f(x, y)$ real and even	$\Leftrightarrow F(u, v)$ real and even
9)	$f(x, y)$ real and odd	$\Leftrightarrow F(u, v)$ imaginary and odd
10)	$f(x, y)$ imaginary and even	$\Leftrightarrow F(u, v)$ imaginary and even
11)	$f(x, y)$ imaginary and odd	$\Leftrightarrow F(u, v)$ real and odd
12)	$f(x, y)$ complex and even	$\Leftrightarrow F(u, v)$ complex and even
13)	$f(x, y)$ complex and odd	$\Leftrightarrow F(u, v)$ complex and odd

TABLE 4.1 Some symmetry properties of the 2-D DFT and its inverse. $R(u, v)$ and $I(u, v)$ are the real and imaginary parts of $F(u, v)$, respectively. The term *complex* indicates that a function has nonzero real and imaginary parts.

[†]Recall that x, y, u , and v are *discrete* (integer) variables, with x and u in the range $[0, M - 1]$, and y , and v in the range $[0, N - 1]$. To say that a complex function is *even* means that its real *and* imaginary parts are even, and similarly for an odd complex function.

Properties of the 2-D DFT

Fourier Spectrum and Phase Angle

2-D DFT in polar form

$$F(u, v) = |F(u, v)| e^{j\phi(u, v)}$$

Fourier spectrum

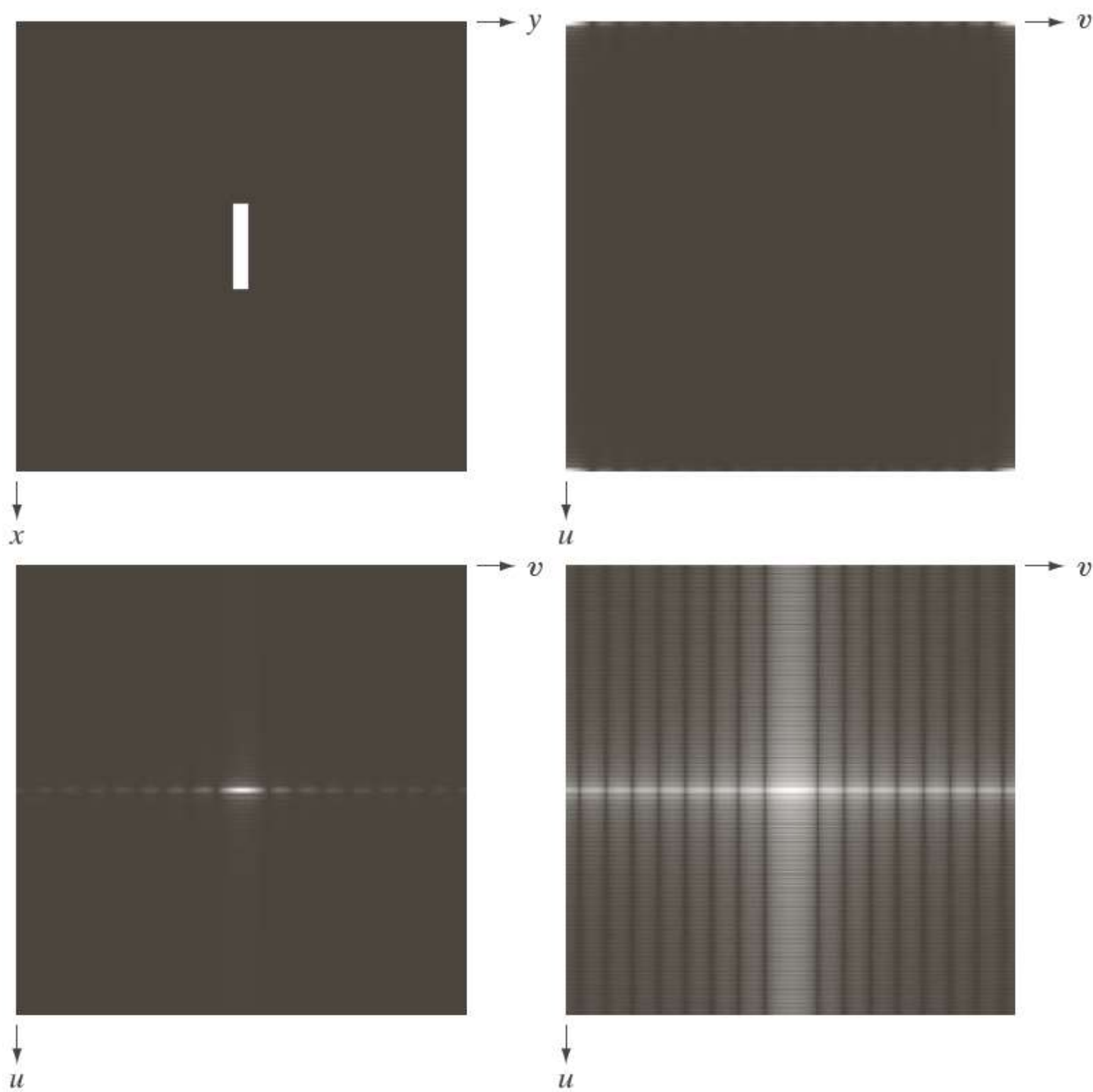
$$|F(u, v)| = \left[R^2(u, v) + I^2(u, v) \right]^{1/2}$$

Power spectrum

$$P(u, v) = |F(u, v)|^2 = R^2(u, v) + I^2(u, v)$$

Phase angle

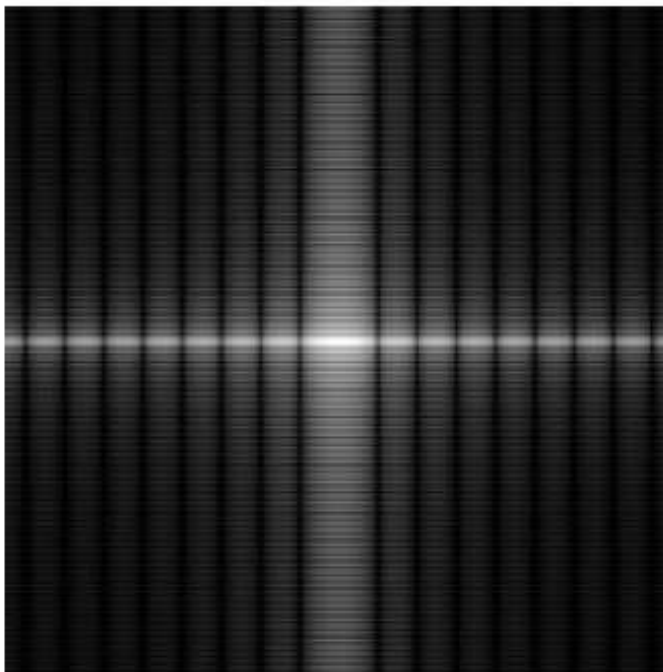
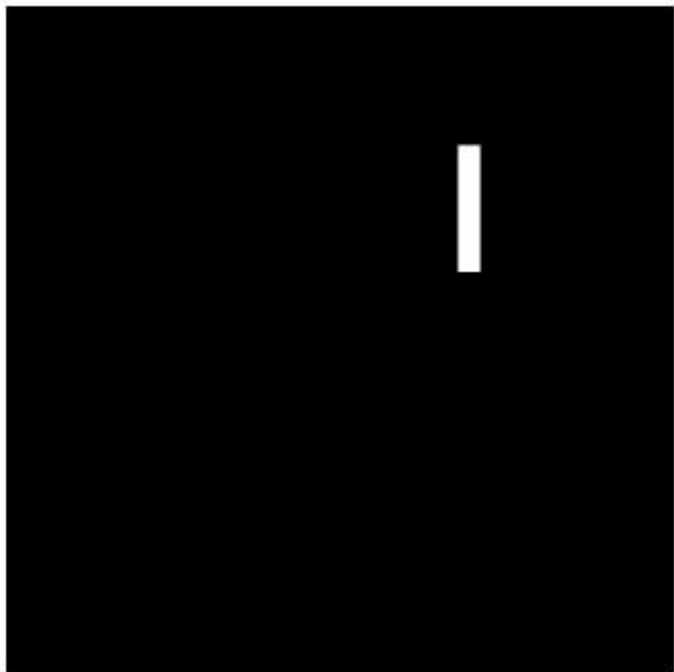
$$\phi(u, v) = \arctan \left[\frac{I(u, v)}{R(u, v)} \right]$$



a	b
c	d

FIGURE 4.24

(a) Image. (b) Spectrum showing bright spots in the four corners. (c) Centered spectrum. (d) Result showing increased detail after a log transformation. The zero crossings of the spectrum are closer in the vertical direction because the rectangle in (a) is longer in that direction. The coordinate convention used throughout the book places the origin of the spatial and frequency domains at the top left.

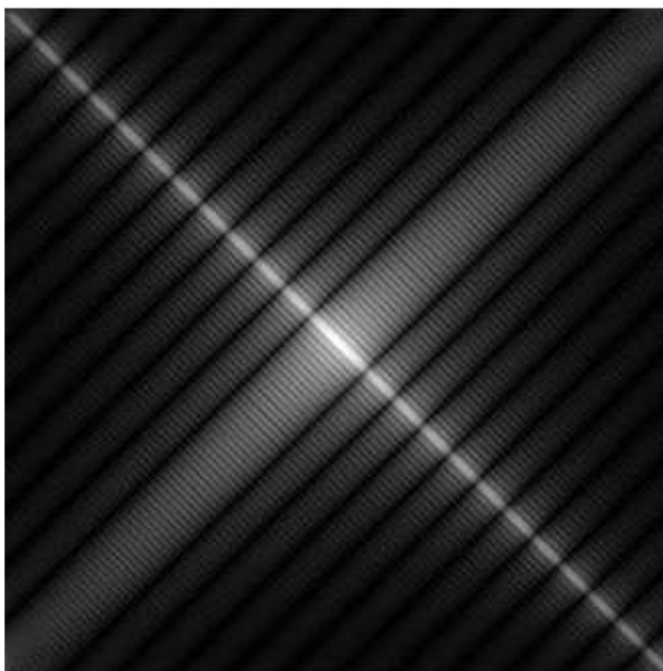
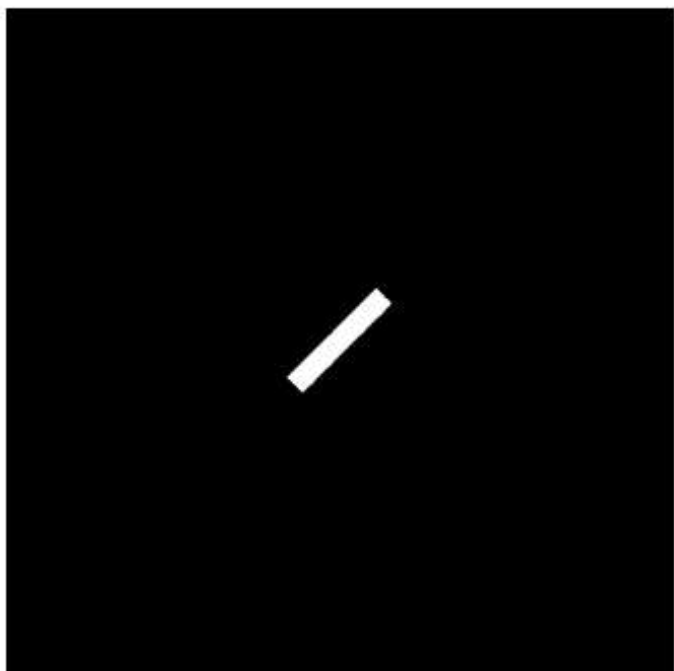


a	b
c	d

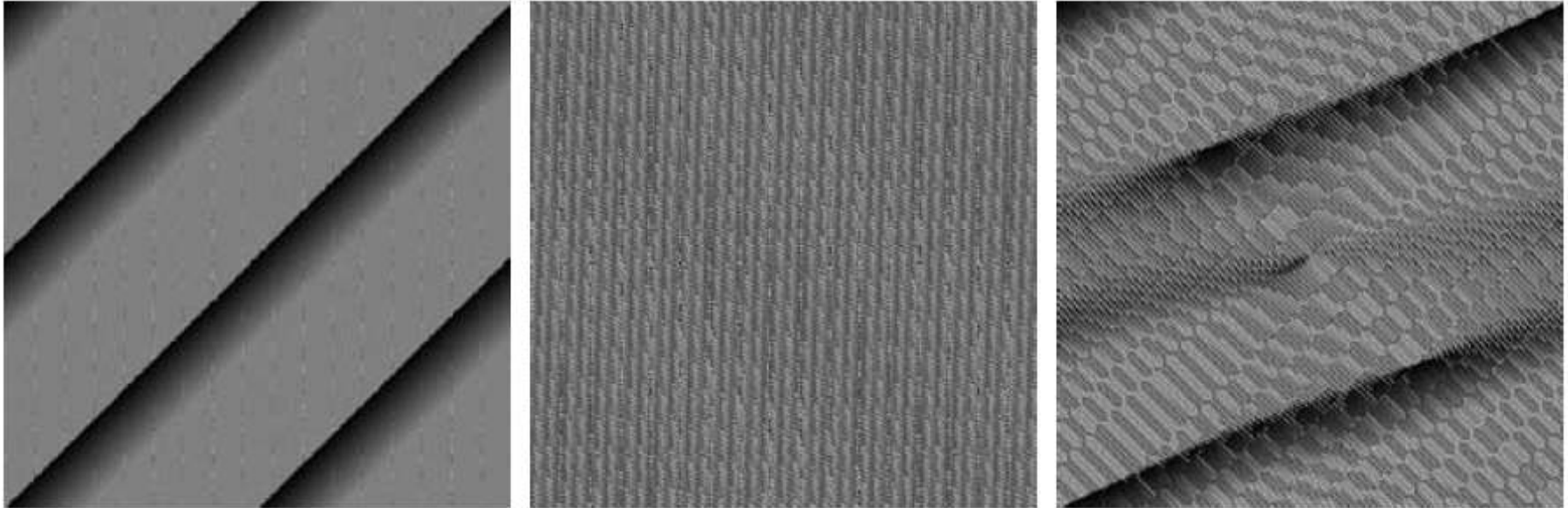
FIGURE 4.25

(a) The rectangle in Fig. 4.24(a) translated, and (b) the corresponding spectrum.

(c) Rotated rectangle, and (d) the corresponding spectrum. The spectrum corresponding to the translated rectangle is identical to the spectrum corresponding to the original image in Fig. 4.24(a).



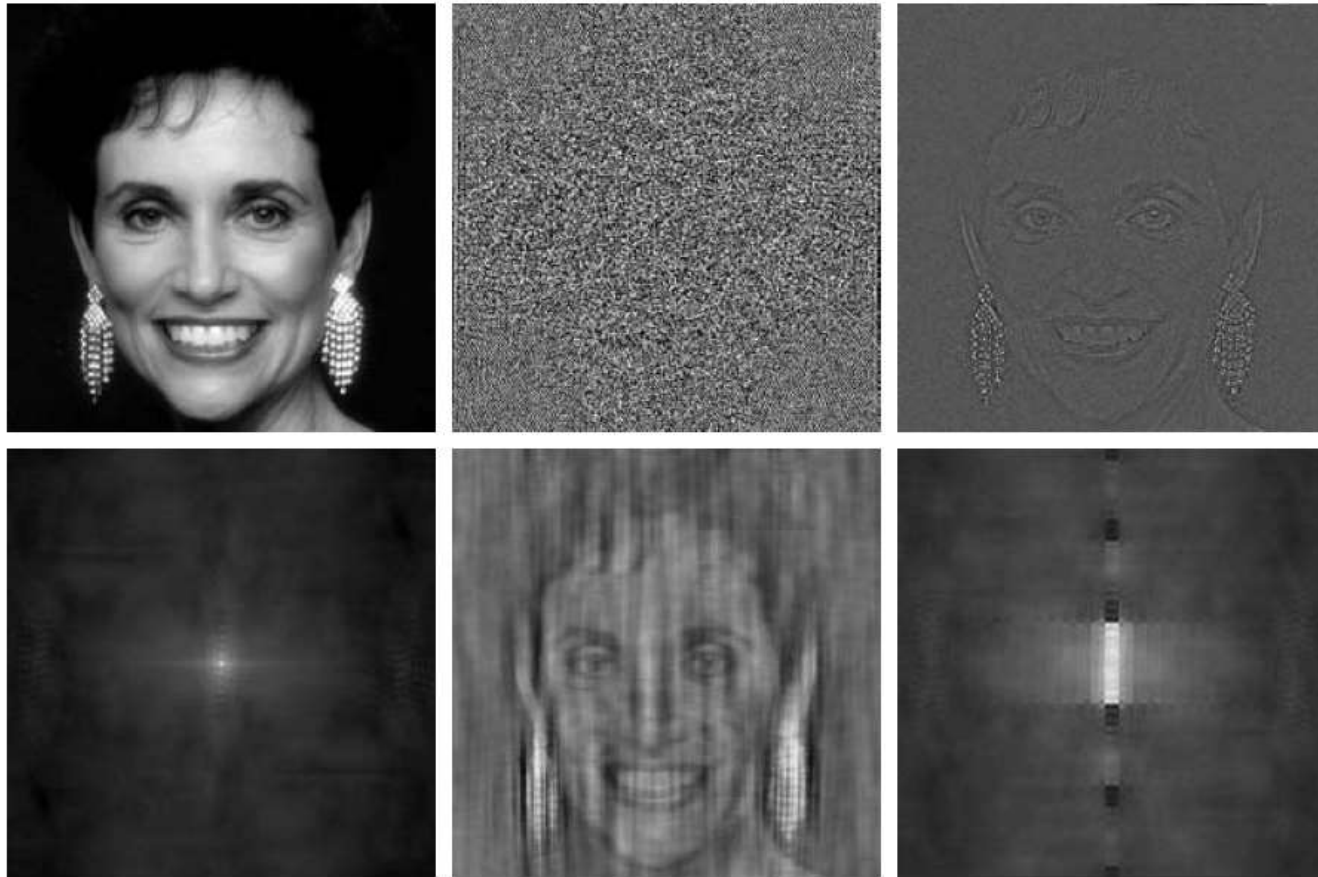
Example: Phase Angles



a b c

FIGURE 4.26 Phase angle array corresponding (a) to the image of the centered rectangle in Fig. 4.24(a), (b) to the translated image in Fig. 4.25(a), and (c) to the rotated image in Fig. 4.25(c).

Example: Phase Angles and The Reconstructu



a	b	c
d	e	f

FIGURE 4.27 (a) Woman. (b) Phase angle. (c) Woman reconstructed using only the phase angle. (d) Woman reconstructed using only the spectrum. (e) Reconstruction using the phase angle corresponding to the woman and the spectrum corresponding to the rectangle in Fig. 4.24(a). (f) Reconstruction using the phase of the rectangle and the spectrum of the woman.

2-D Convolution Theorem

1-D convolution

$$f(x) \star h(x) = \sum_{m=0}^{M-1} f(m)h(x-m)$$

2-D convolution

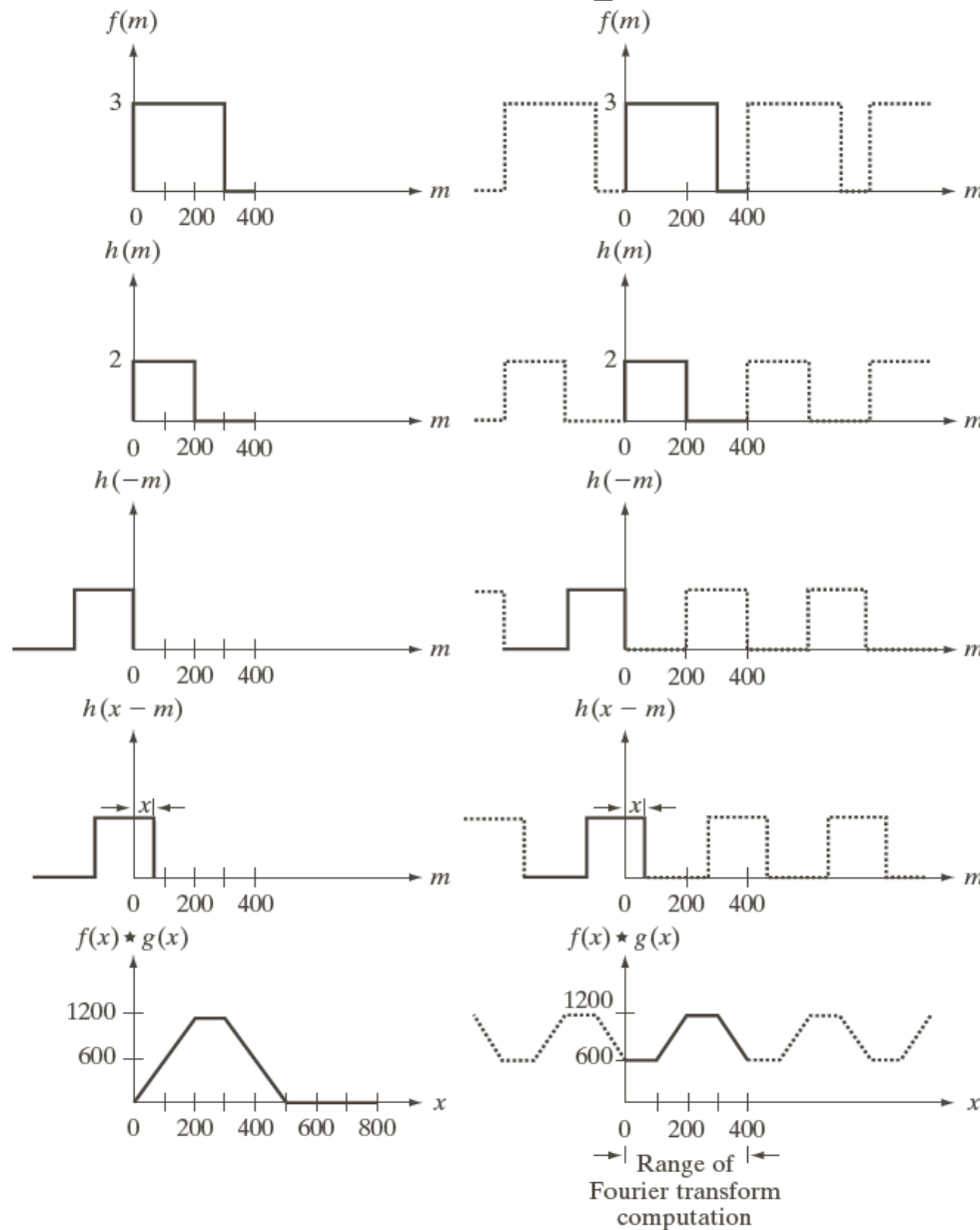
$$f(x, y) \star h(x, y) = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} f(m, n)h(x-m, y-n)$$

$$x = 0, 1, 2, \dots, M-1; y = 0, 1, 2, \dots, N-1.$$

$$f(x, y) \star h(x, y) \Leftrightarrow F(u, v)H(u, v)$$

$$f(x, y)h(x, y) \Leftrightarrow F(u, v) \star H(u, v)$$

An Example of Convolution

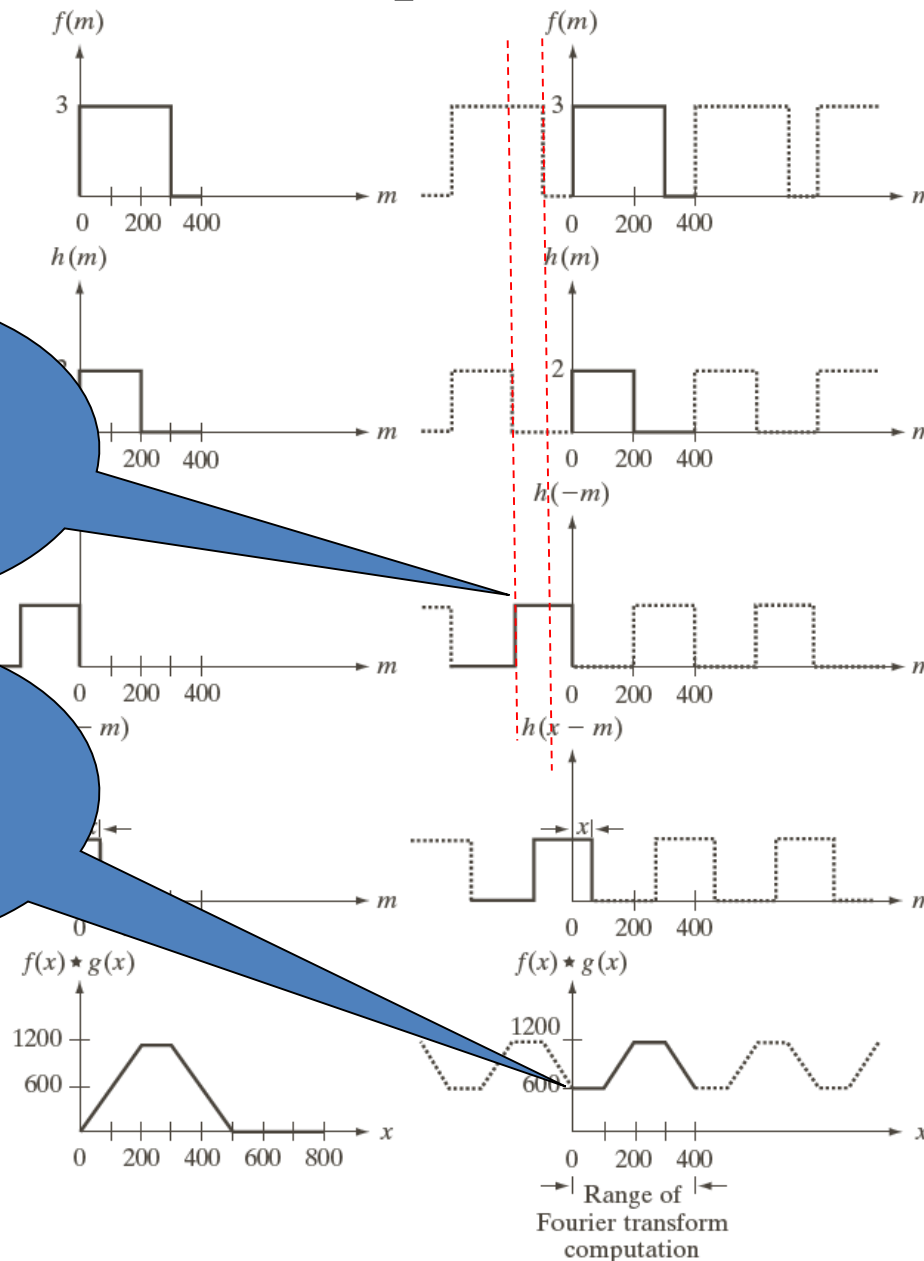


Mirroring h
about the origin

Translating the
mirrored
function by x

Computing the
sum for each x

An Example of Convolution



a	f
b	g
c	h
d	i
e	j

FIGURE 4.28 Left column: convolution of two discrete functions obtained using the approach discussed in Section 3.4.2. The result in (e) is correct. Right column: Convolution of the same functions, but taking into account the periodicity implied by the DFT. Note in (j) how data from adjacent periods produce wraparound error, yielding an incorrect convolution result. To obtain the correct result, function padding must be used.

Zero Padding

- ◆ Consider two functions $f(x)$ and $h(x)$ composed of A and B samples, respectively
- ◆ Append zeros to both functions so that they have the same length, denoted by P , then wraparound is avoided by choosing

$$P \geq A+B-1$$

Zero Padding

- ◆ Let $f(x,y)$ and $h(x,y)$ be two image arrays of sizes $A \times B$ and $C \times D$ pixels, respectively. Wraparound error in their convolution can be avoided by padding these functions with zeros

$$f_p(x, y) = \begin{cases} f(x, y) & 0 \leq x \leq A-1 \text{ and } 0 \leq y \leq B-1 \\ 0 & A \leq x \leq P \text{ or } B \leq y \leq Q \end{cases}$$

$$h_p(x, y) = \begin{cases} h(x, y) & 0 \leq x \leq C-1 \text{ and } 0 \leq y \leq D-1 \\ 0 & C \leq x \leq P \text{ or } D \leq y \leq Q \end{cases}$$

Here $P \geq A + C - 1; Q \geq B + D - 1$

Summary

Name	Expression(s)
1) Discrete Fourier transform (DFT) of $f(x, y)$	$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi(ux/M+vy/N)}$
2) Inverse discrete Fourier transform (IDFT) of $F(u, v)$	$f(x, y) = \frac{1}{MN} \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u, v) e^{j2\pi(ux/M+vy/N)}$
3) Polar representation	$F(u, v) = F(u, v) e^{j\phi(u, v)}$
4) Spectrum	$ F(u, v) = [R^2(u, v) + I^2(u, v)]^{1/2}$ $R = \text{Real}(F); \quad I = \text{Imag}(F)$
5) Phase angle	$\phi(u, v) = \tan^{-1} \left[\frac{I(u, v)}{R(u, v)} \right]$
6) Power spectrum	$P(u, v) = F(u, v) ^2$
7) Average value	$\bar{f}(x, y) = \frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) = \frac{1}{MN} F(0, 0)$

(Continued)

Summary

Name	Expression(s)
8) Periodicity (k_1 and k_2 are integers)	$F(u, v) = F(u + k_1M, v) = F(u, v + k_2N)$ $= F(u + k_1M, v + k_2N)$ $f(x, y) = f(x + k_1M, y) = f(x, y + k_2N)$ $= f(x + k_1M, y + k_2N)$
9) Convolution	$f(x, y) \star h(x, y) = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} f(m, n)h(x - m, y - n)$
10) Correlation	$f(x, y) \star h(x, y) = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} f^*(m, n)h(x + m, y + n)$
11) Separability	The 2-D DFT can be computed by computing 1-D DFT transforms along the rows (columns) of the image, followed by 1-D transforms along the columns (rows) of the result. See Section 4.11.1.
12) Obtaining the inverse Fourier transform using a forward transform algorithm.	$MNf^*(x, y) = \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F^*(u, v)e^{-j2\pi(ux/M+vy/N)}$ This equation indicates that inputting $F^*(u, v)$ into an algorithm that computes the forward transform (right side of above equation) yields $MNf^*(x, y)$. Taking the complex conjugate and dividing by MN gives the desired inverse. See Section 4.11.2.

Summary

Name	DFT Pairs
1) Symmetry properties	See Table 4.1
2) Linearity	$af_1(x, y) + bf_2(x, y) \Leftrightarrow aF_1(u, v) + bF_2(u, v)$
3) Translation (general)	$f(x, y)e^{j2\pi(u_0x/M+v_0y/N)} \Leftrightarrow F(u - u_0, v - v_0)$ $f(x - x_0, y - y_0) \Leftrightarrow F(u, v)e^{-j2\pi(ux_0/M+vy_0/N)}$
4) Translation to center of the frequency rectangle, $(M/2, N/2)$	$f(x, y)(-1)^{x+y} \Leftrightarrow F(u - M/2, v - N/2)$ $f(x - M/2, y - N/2) \Leftrightarrow F(u, v)(-1)^{u+v}$
5) Rotation	$f(r, \theta + \theta_0) \Leftrightarrow F(\omega, \varphi + \theta_0)$ $x = r \cos \theta \quad y = r \sin \theta \quad u = \omega \cos \varphi \quad v = \omega \sin \varphi$
6) Convolution theorem [†]	$f(x, y) \star h(x, y) \Leftrightarrow F(u, v)H(u, v)$ $f(x, y)h(x, y) \Leftrightarrow F(u, v) \star H(u, v)$

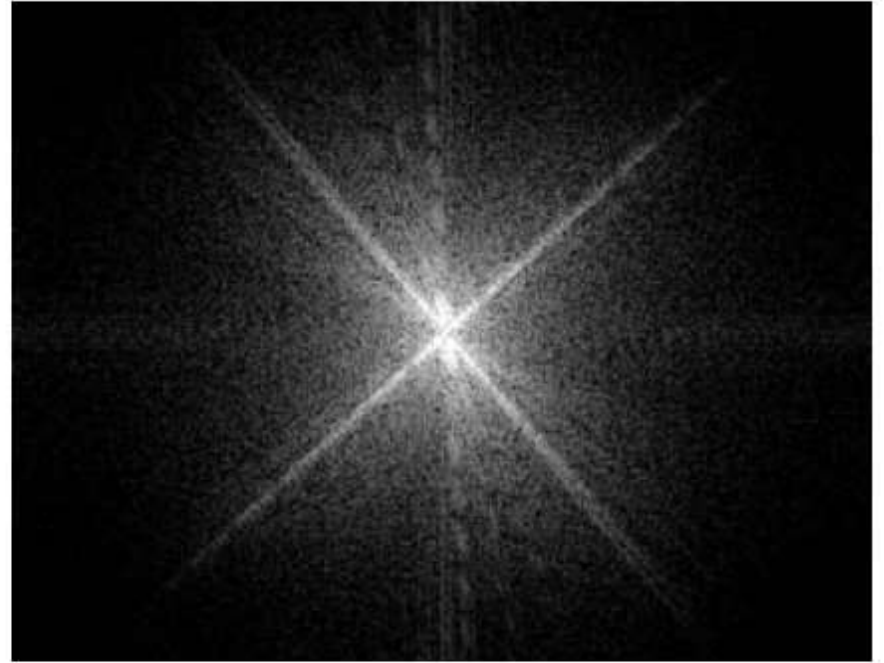
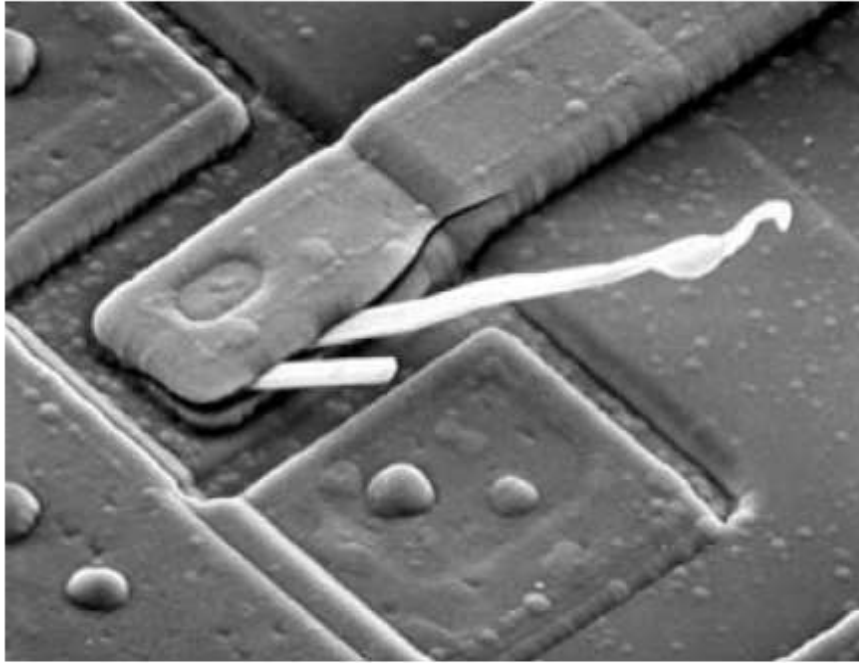
(Continued)

Summary

Name	DFT Pairs
7) Correlation theorem [†]	$f(x, y) \star h(x, y) \Leftrightarrow F^*(u, v) H(u, v)$ $f^*(x, y) h(x, y) \Leftrightarrow F(u, v) \star H(u, v)$
8) Discrete unit impulse	$\delta(x, y) \Leftrightarrow 1$
9) Rectangle	$\text{rect}[a, b] \Leftrightarrow ab \frac{\sin(\pi ua)}{(\pi ua)} \frac{\sin(\pi vb)}{(\pi vb)} e^{-j\pi(ua+vb)}$
10) Sine	$\sin(2\pi u_0 x + 2\pi v_0 y) \Leftrightarrow$ $j \frac{1}{2} [\delta(u + Mu_0, v + Nv_0) - \delta(u - Mu_0, v - Nv_0)]$
11) Cosine	$\cos(2\pi u_0 x + 2\pi v_0 y) \Leftrightarrow$ $\frac{1}{2} [\delta(u + Mu_0, v + Nv_0) + \delta(u - Mu_0, v - Nv_0)]$
The following Fourier transform pairs are derivable only for continuous variables, denoted as before by t and z for spatial variables and by μ and ν for frequency variables. These results can be used for DFT work by sampling the continuous forms.	
12) <i>Differentiation</i> (The expressions on the right assume that $f(\pm\infty, \pm\infty) = 0$.)	$\left(\frac{\partial}{\partial t}\right)^m \left(\frac{\partial}{\partial z}\right)^n f(t, z) \Leftrightarrow (j2\pi\mu)^m (j2\pi\nu)^n F(\mu, \nu)$ $\frac{\partial^m f(t, z)}{\partial t^m} \Leftrightarrow (j2\pi\mu)^m F(\mu, \nu); \frac{\partial^n f(t, z)}{\partial z^n} \Leftrightarrow (j2\pi\nu)^n F(\mu, \nu)$
13) <i>Gaussian</i>	$A 2\pi\sigma^2 e^{-2\pi^2\sigma^2(t^2+z^2)} \Leftrightarrow A e^{-(\mu^2+\nu^2)/2\sigma^2}$ (A is a constant)

[†] Assumes that the functions have been extended by zero padding. Convolution and correlation are associative, commutative, and distributive.

Basic Filtering in the Frequency Domain



a b

FIGURE 4.29 (a) SEM image of a damaged integrated circuit. (b) Fourier spectrum of (a). (Original image courtesy of Dr. J. M. Hudak, Brockhouse Institute for Materials Research, McMaster University, Hamilton, Ontario, Canada.)

The Basic Filtering in the Frequency Domain

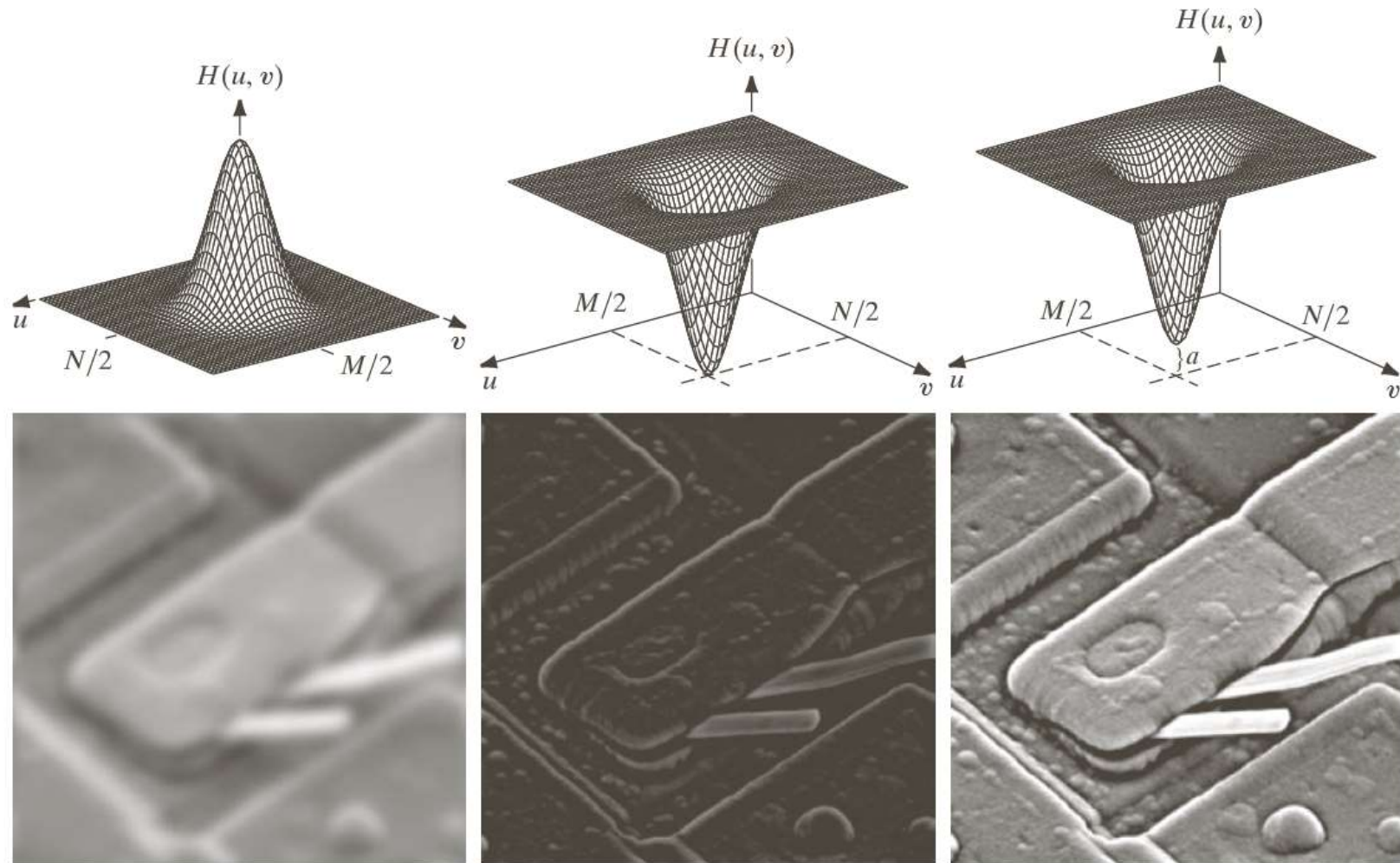
- ◆ Modifying the Fourier transform of an image
- ◆ Computing the inverse transform to obtain the processed result

$$g(x, y) = \mathfrak{F}^{-1}\{H(u, v)F(u, v)\}$$

$F(u, v)$ is the DFT of the input image

$H(u, v)$ is a filter function.

The Basic Filtering in the Frequency Domain



a	b	c
d	e	f

FIGURE 4.31 Top row: frequency domain filters. Bottom row: corresponding filtered images obtained using Eq. (4.7-1). We used $a = 0.85$ in (c) to obtain (f) (the height of the filter itself is 1). Compare (f) with Fig. 4.29(a).

The Basic Filtering in the Frequency Domain

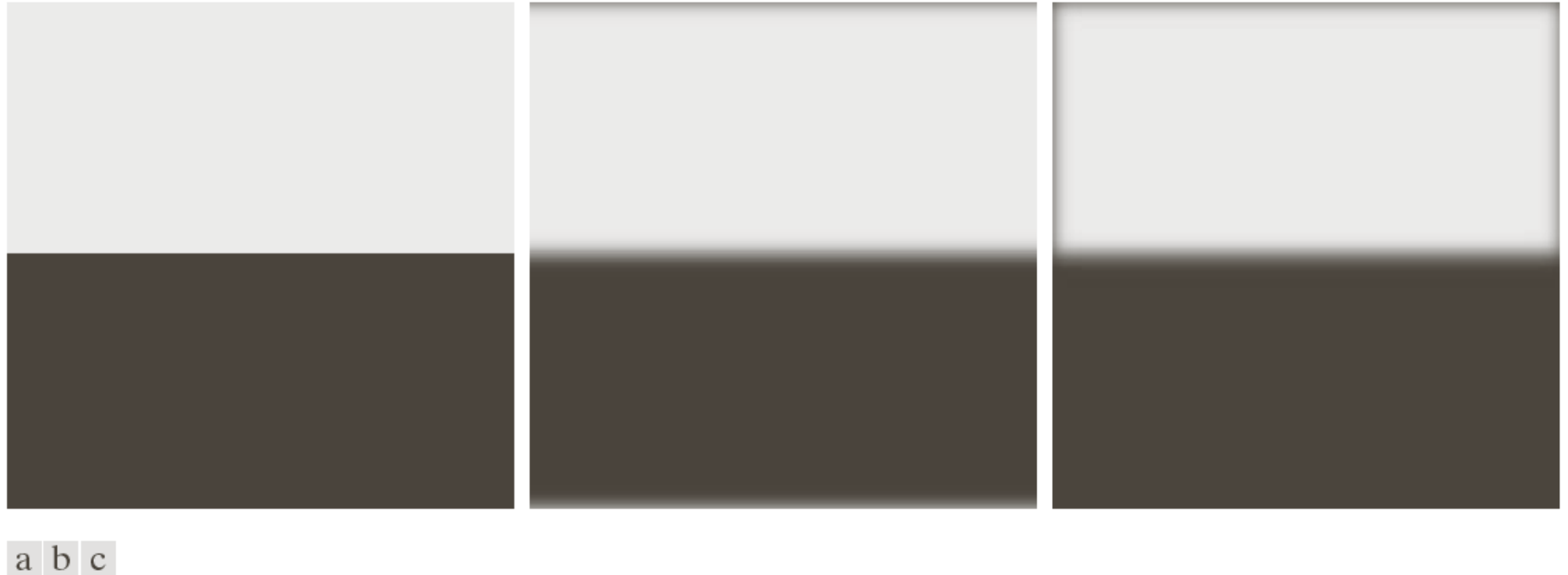


FIGURE 4.32 (a) A simple image. (b) Result of blurring with a Gaussian lowpass filter without padding. (c) Result of lowpass filtering with padding. Compare the light area of the vertical edges in (b) and (c).

Zero-Phase-Shift Filters

$$g(x, y) = \mathfrak{F}^{-1}\{H(u, v)F(u, v)\}$$

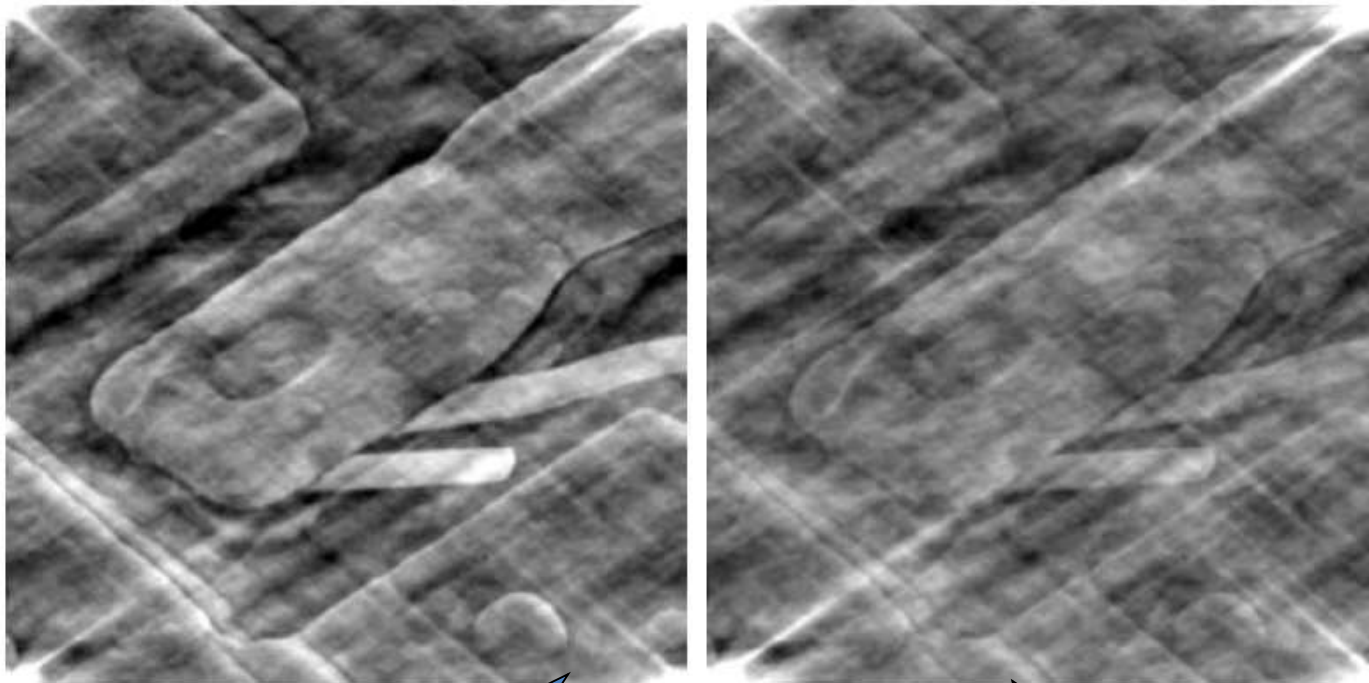
$$F(u, v) = R(u, v) + jI(u, v)$$

$$g(x, y) = \mathfrak{F}^{-1}[H(u, v)R(u, v) + jH(u, v)I(u, v)]$$

Filters affect the real and imaginary parts equally,
and thus no effect on the phase.

These filters are called **zero-phase-shift** filters

Examples: Nonzero-Phase-Shift Filters



a b

FIGURE 4.35

(a) Image resulting from multiplying by 0.5 the phase angle in Eq. (4.6-15) and then computing the IDFT. (b) The result of multiplying the phase by 0.25. The spectrum was not changed in either of the two cases.

Phase angle is
multiplied by
0.5

Phase angle is
multiplied by
0.5

Even small changes in the phase angle can have dramatic (usually undesirable) effects on the filtered output

Summary:

Steps for Filtering in the Frequency Domain

1. Given an input image $f(x,y)$ of size $M \times N$, obtain the padding parameters P and Q . Typically, $P = 2M$ and $Q = 2N$.
2. Form a padded image, $f_p(x,y)$ of size $P \times Q$ by appending the necessary number of zeros to $f(x,y)$
3. Multiply $f_p(x,y)$ by $(-1)^{x+y}$ to center its transform
4. Compute the DFT, $F(u,v)$ of the image from step 3
5. Generate a real, symmetric filter function*, $H(u,v)$, of size $P \times Q$ with center at coordinates $(P/2, Q/2)$

*generate from a given spatial filter, we pad the spatial filter, multiply the expanded array by $(-1)^{x+y}$, and compute the DFT of the result to obtain a centered $H(u,v)$.

Summary:

Steps for Filtering in the Frequency Domain

6. Form the product $G(u,v) = H(u,v)F(u,v)$ using array multiplication

7. Obtain the processed image

$$g_p(x, y) = \left\{ \text{real} \left[\mathfrak{F}^{-1} [G(u, v)] \right] \right\} (-1)^{x+y}$$

8. Obtain the final processed result, $g(x,y)$, by extracting the $M \times N$ region from the top, left quadrant of $g_p(x,y)$

Correspondence Between Filtering in the Spatial and Frequency Domains (1)

Let $H(u)$ denote the 1-D frequency domain Gaussian filter

$$H(u) = Ae^{-u^2/2\sigma^2}$$

The corresponding filter in the spatial domain

$$h(x) = \sqrt{2\pi}\sigma Ae^{-2\pi^2\sigma^2x^2}$$

1. Both components are Gaussian and real
2. The functions behave reciprocally

Correspondence Between Filtering in the Spatial and Frequency Domains (2)

Let $H(u)$ denote the difference of Gaussian filter

$$H(u) = Ae^{-u^2/2\sigma_1^2} - Be^{-u^2/2\sigma_2^2}$$

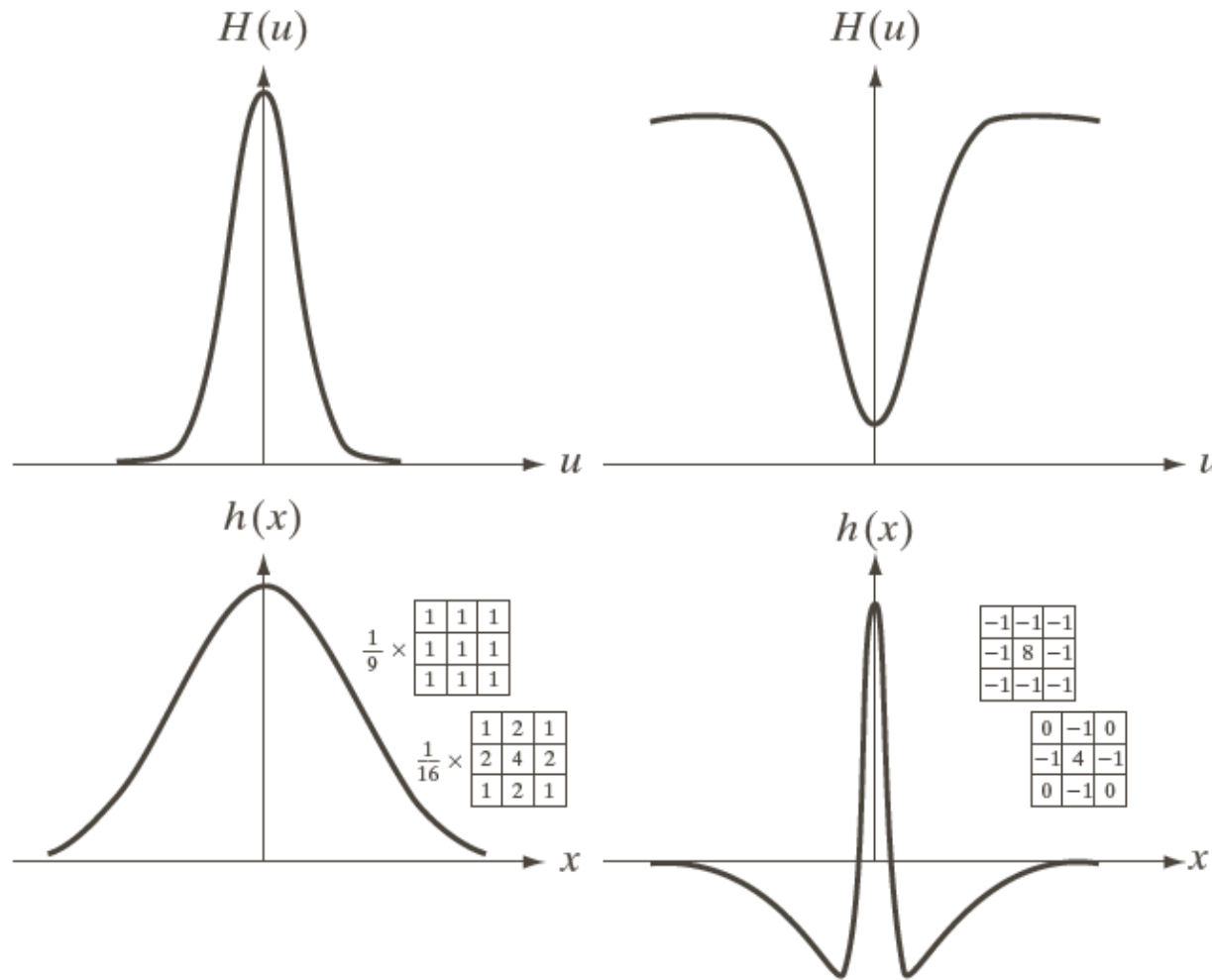
with $A \geq B$ and $\sigma_1 \geq \sigma_2$

The corresponding filter in the spatial domain

$$h(x) = \sqrt{2\pi}\sigma_1 Ae^{-2\pi^2\sigma_1^2x^2} - \sqrt{2\pi}\sigma_2 Ae^{-2\pi^2\sigma_2^2x^2}$$

High-pass filter or low-pass filter ?

Correspondence Between Filtering in the Spatial and Frequency Domains (3)



a	c
b	d

FIGURE 4.37

(a) A 1-D Gaussian lowpass filter in the frequency domain.

(b) Spatial lowpass filter corresponding to (a).

(c) Gaussian highpass filter in the frequency domain.

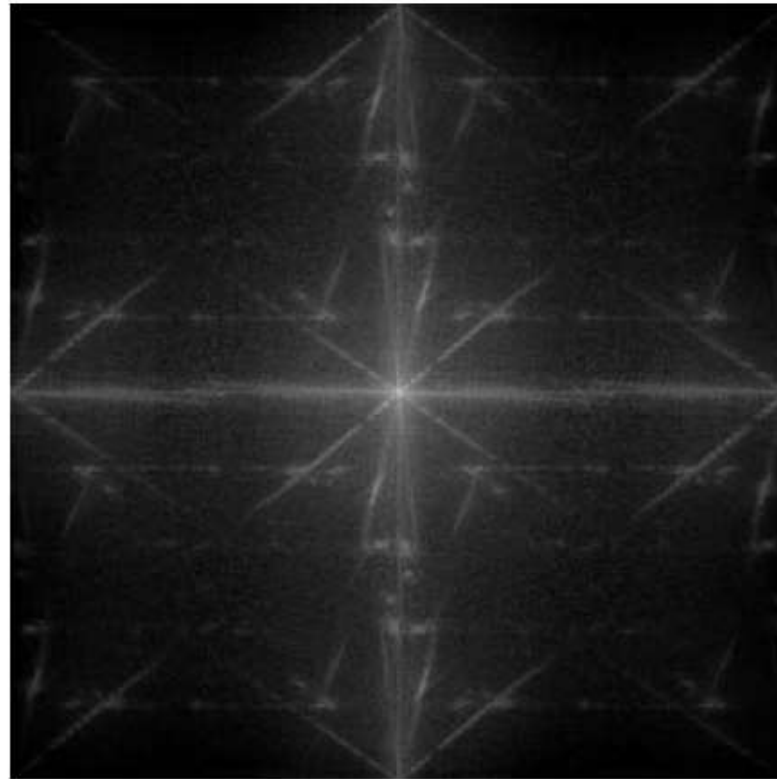
(d) Spatial highpass filter corresponding to (c).

The small 2-D masks shown are spatial filters we used in Chapter 3.

Correspondence Between Filtering in the Spatial and Frequency Domains: Example



600x600

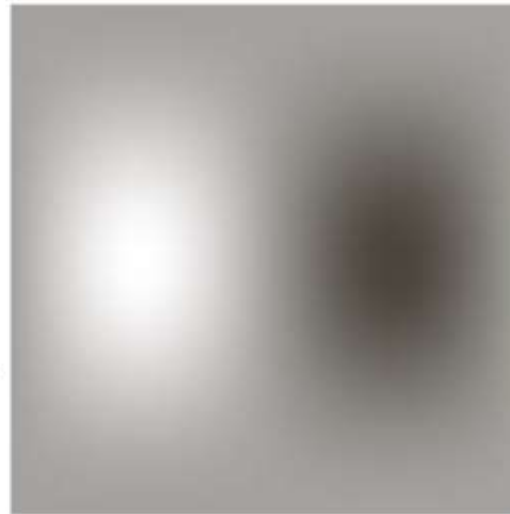
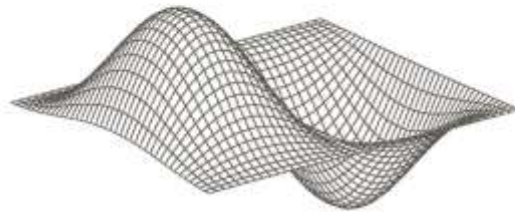


a b

FIGURE 4.38
(a) Image of a building, and
(b) its spectrum.

Correspondence Between Filtering in the Spatial and Frequency Domains: Example

-1	0	1
-2	0	2
-1	0	1



a	b
c	d

FIGURE 4.39

(a) A spatial mask and perspective plot of its corresponding frequency domain filter. (b) Filter shown as an image. (c) Result of filtering Fig. 4.38(a) in the frequency domain with the filter in (b). (d) Result of filtering the same image with the spatial filter in (a). The results are identical.



Generate H(u,v)

$$f_p(x, y) = \begin{cases} f(x, y) & 0 \leq x \leq 599 \text{ and } 0 \leq y \leq 599 \\ 0 & 600 \leq x \leq 602 \text{ or } 600 \leq y \leq 602 \end{cases}$$

$$h_p(x, y) = \begin{cases} h(x, y) & 0 \leq x \leq 2 \text{ and } 0 \leq y \leq 2 \\ 0 & 3 \leq x \leq 602 \text{ or } 3 \leq y \leq 602 \end{cases}$$

Here $P \geq A(600) + C(3) - 1 = 602$;

$Q \geq B(600) + D(3) - 1 = 602$.

Generate $H(u,v)$

1. Multiply $h_p(x, y)$ by $(-1)^{x+y}$ to center the frequency domain filter
2. Compute the forward DFT of the result in (1)
3. Set the real part of the resulting DFT to 0 to account for parasitic real parts
4. Multiply the result by $(-1)^{u+v}$, which is implicit when $h(x, y)$ was moved to the center of $h_p(x, y)$.

Image Smoothing Using Frequency Domain Filters: ILPF

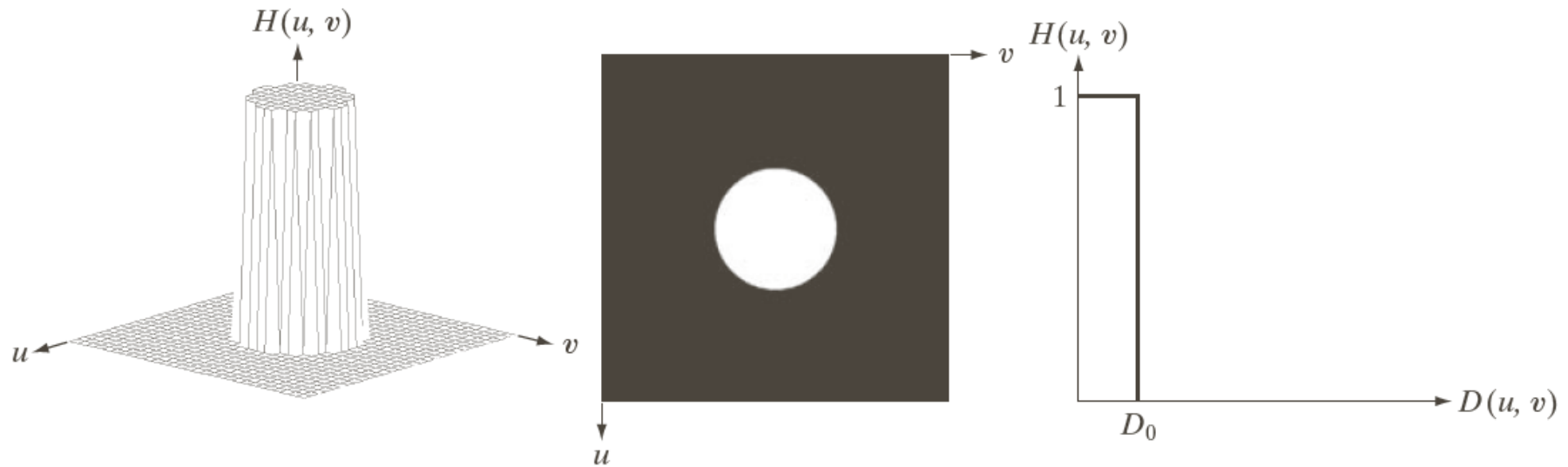
Ideal Lowpass Filters (ILPF)

$$H(u, v) = \begin{cases} 1 & \text{if } D(u, v) \leq D_0 \\ 0 & \text{if } D(u, v) > D_0 \end{cases}$$

D_0 is a positive constant and $D(u, v)$ is the distance between a point (u, v) in the frequency domain and the center of the frequency rectangle

$$D(u, v) = \left[(u - P / 2)^2 + (v - Q / 2)^2 \right]^{1/2}$$

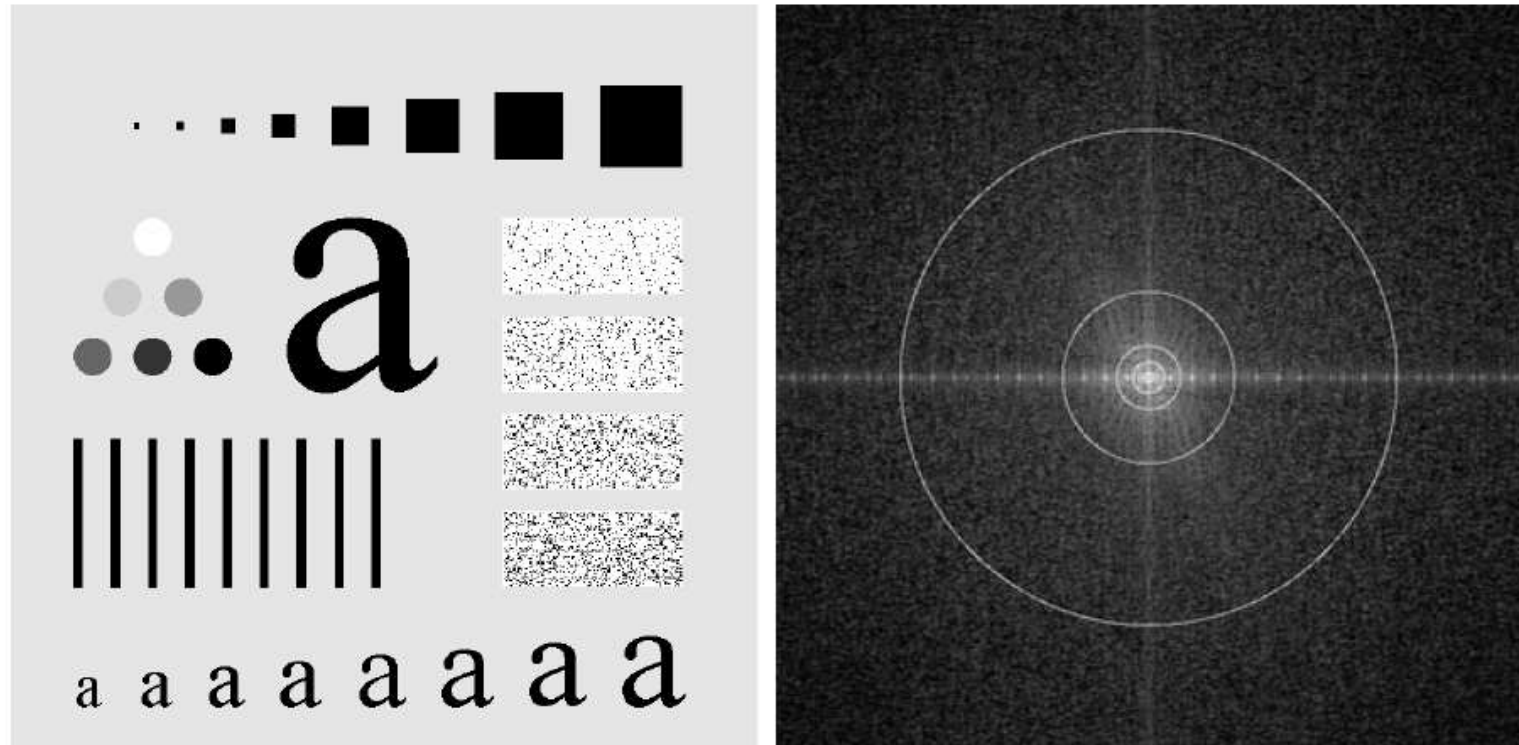
Image Smoothing Using Frequency Domain Filters: ILPF



a b c

FIGURE 4.40 (a) Perspective plot of an ideal lowpass-filter transfer function. (b) Filter displayed as an image. (c) Filter radial cross section.

ILPF Filtering Example



a b

FIGURE 4.41 (a) Test pattern of size 688×688 pixels, and (b) its Fourier spectrum. The spectrum is double the image size due to padding but is shown in half size so that it fits in the page. The superimposed circles have radii equal to 10, 30, 60, 160, and 460 with respect to the full-size spectrum image. These radii enclose 87.0, 93.1, 95.7, 97.8, and 99.2% of the padded image power, respectively.

ILPF Filtering Example

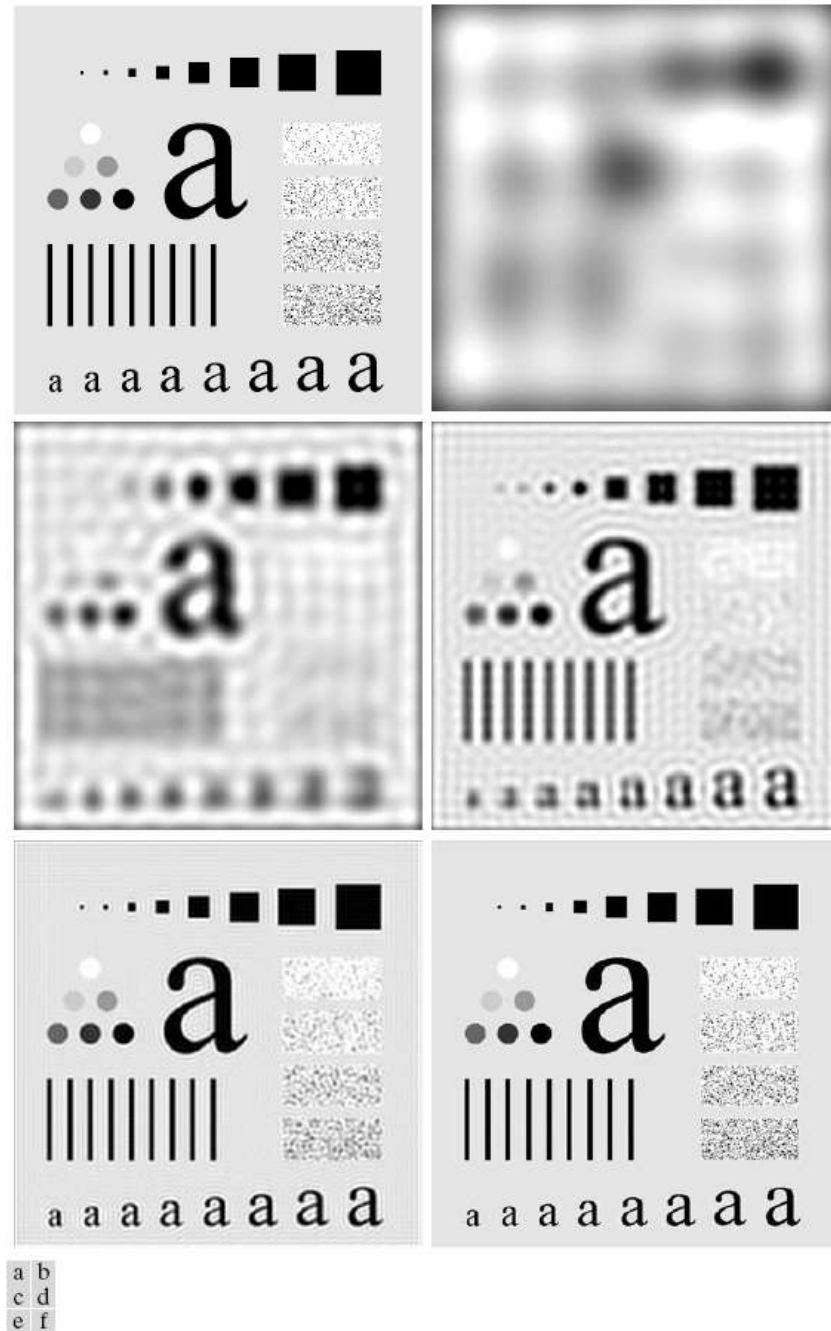
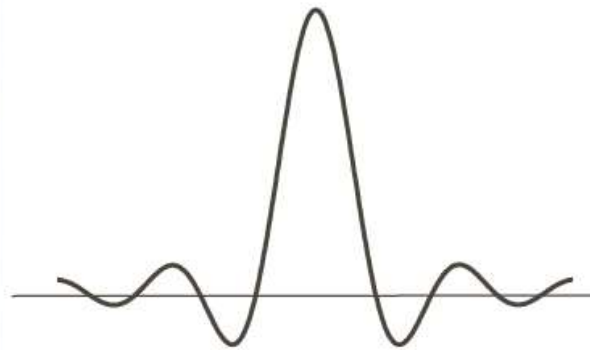
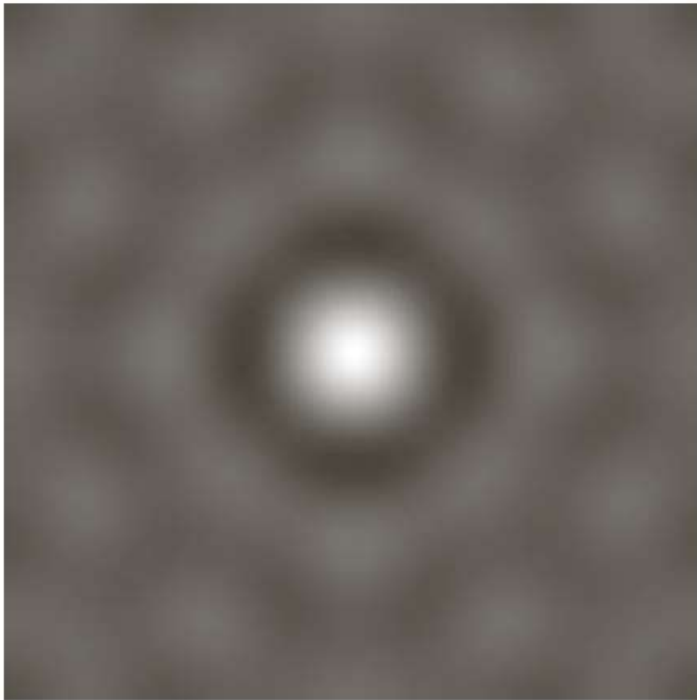


FIGURE 4.42 (a) Original image. (b)–(f) Results of filtering using ILPFs with cutoff frequencies set at radii values 10, 30, 60, 160, and 460, as shown in Fig. 4.41(b). The power removed by these filters was 13, 6.9, 4.3, 2.2, and 0.8% of the total, respectively.

The Spatial Representation of ILPF



a b

FIGURE 4.43

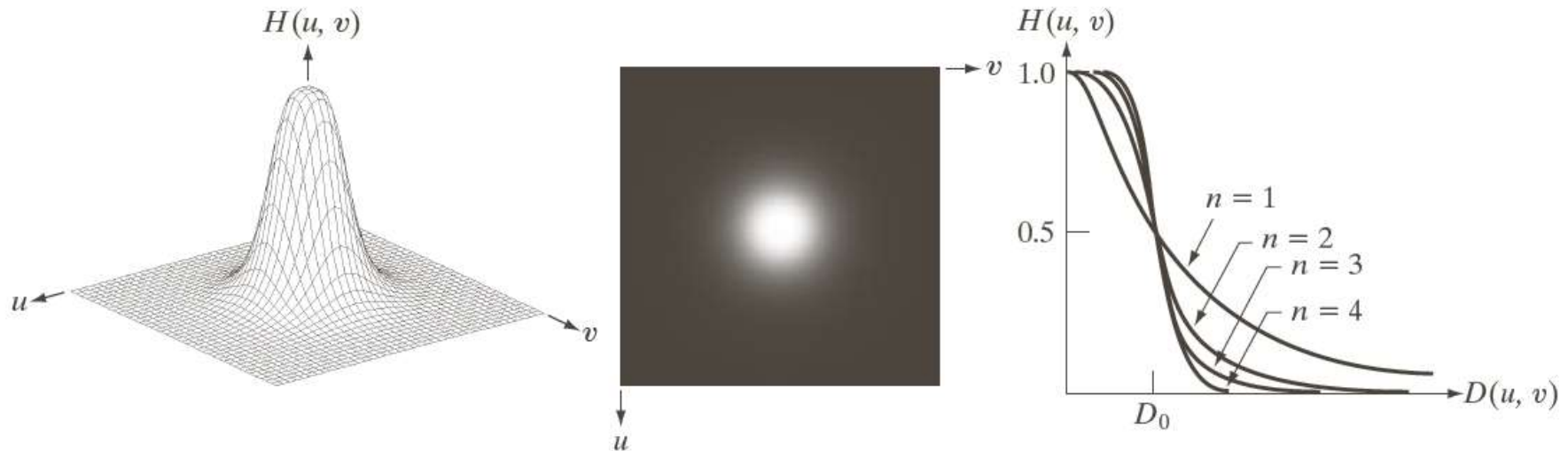
(a) Representation in the spatial domain of an ILPF of radius 5 and size 1000×1000 .

(b) Intensity profile of a horizontal line passing through the center of the image.

Image Smoothing Using Frequency Domain Filters: BLPF

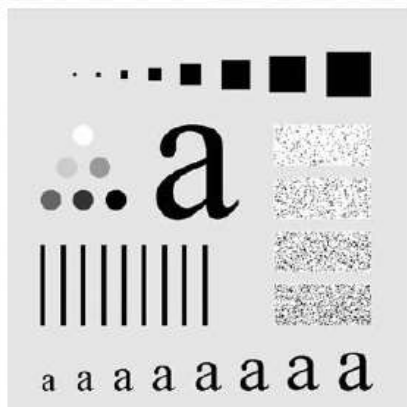
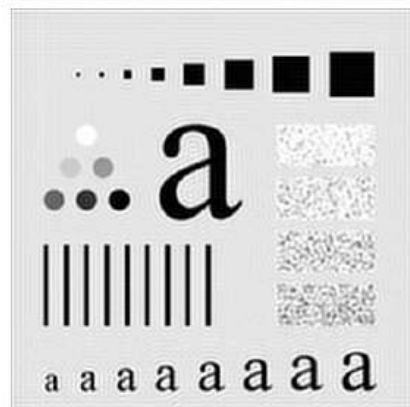
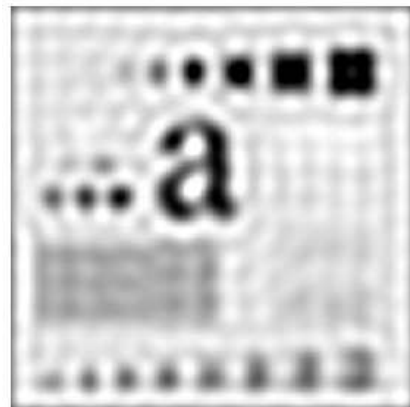
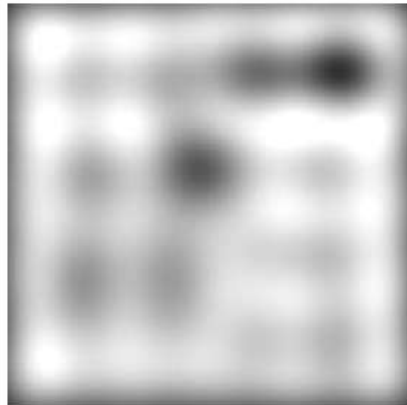
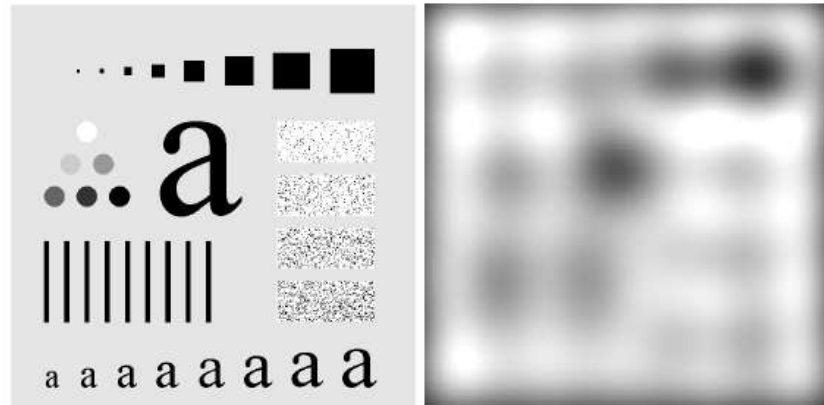
Butterworth Lowpass Filters (BLPF) of order n and with cutoff frequency D_0

$$H(u, v) = \frac{1}{1 + [D(u, v) / D_0]^{2n}}$$



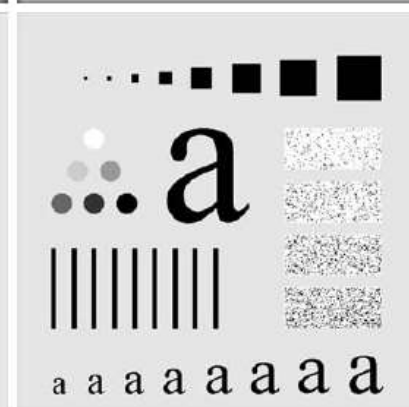
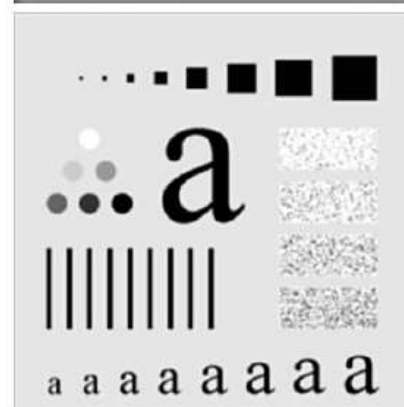
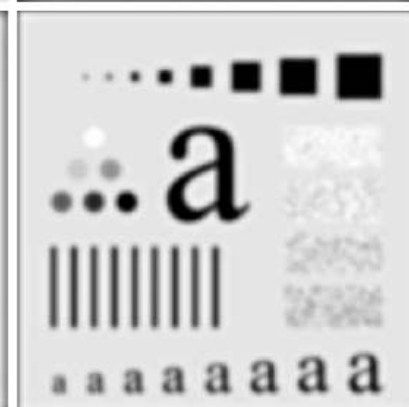
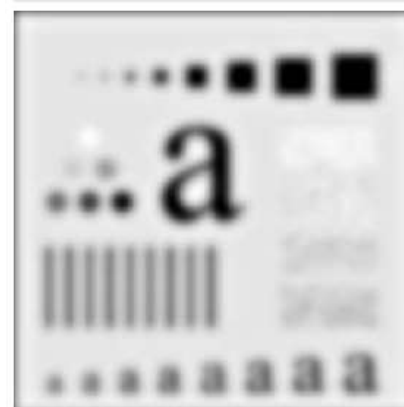
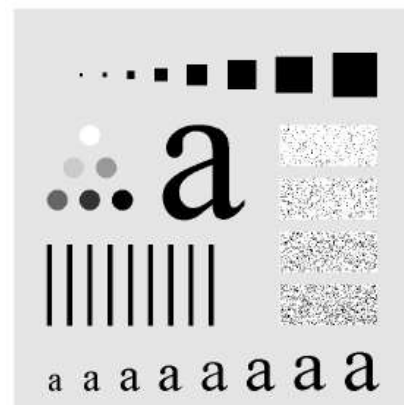
a b c

FIGURE 4.44 (a) Perspective plot of a Butterworth lowpass-filter transfer function. (b) Filter displayed as an image. (c) Filter radial cross sections of orders 1 through 4.



a b
c d
e f

FIGURE 4.42 (a) Original image. (b)–(f) Results of filtering using ILPFs with cutoff frequencies set at radii values 10, 30, 60, 160, and 460, as shown in Fig. 4.41(b). The power removed by these filters was 13, 6.9, 4.3, 2.2, and 0.8% of the total, respectively.



a b
c d
e f

FIGURE 4.45 (a) Original image. (b)–(f) Results of filtering using BLPFs of order 2, with cutoff frequencies at the radii shown in Fig. 4.41. Compare with Fig. 4.42.

The Spatial Representation of BLPF

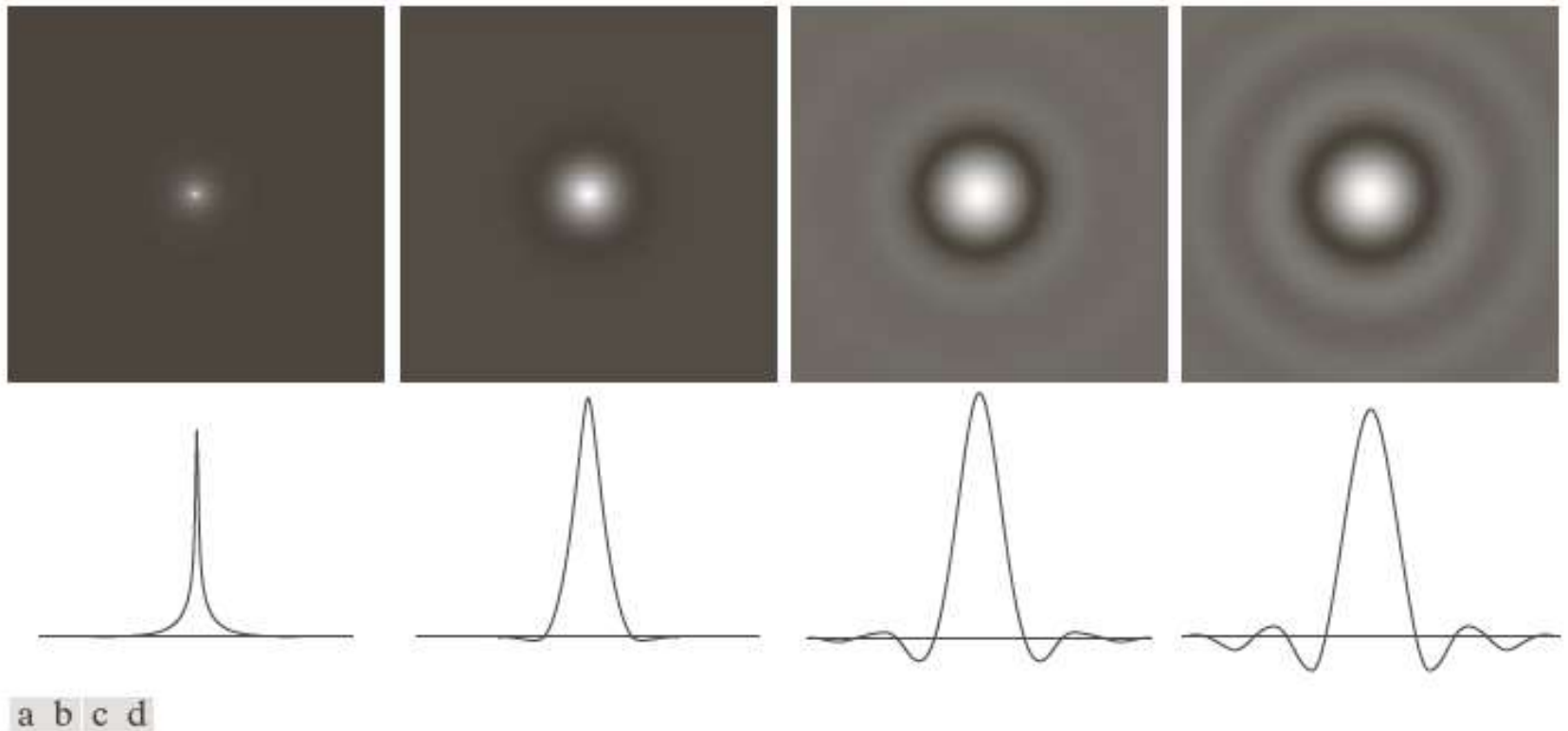


FIGURE 4.46 (a)–(d) Spatial representation of BLPFs of order 1, 2, 5, and 20, and corresponding intensity profiles through the center of the filters (the size in all cases is 1000×1000 and the cutoff frequency is 5). Observe how ringing increases as a function of filter order.

Image Smoothing Using Frequency Domain Filters: GLPF

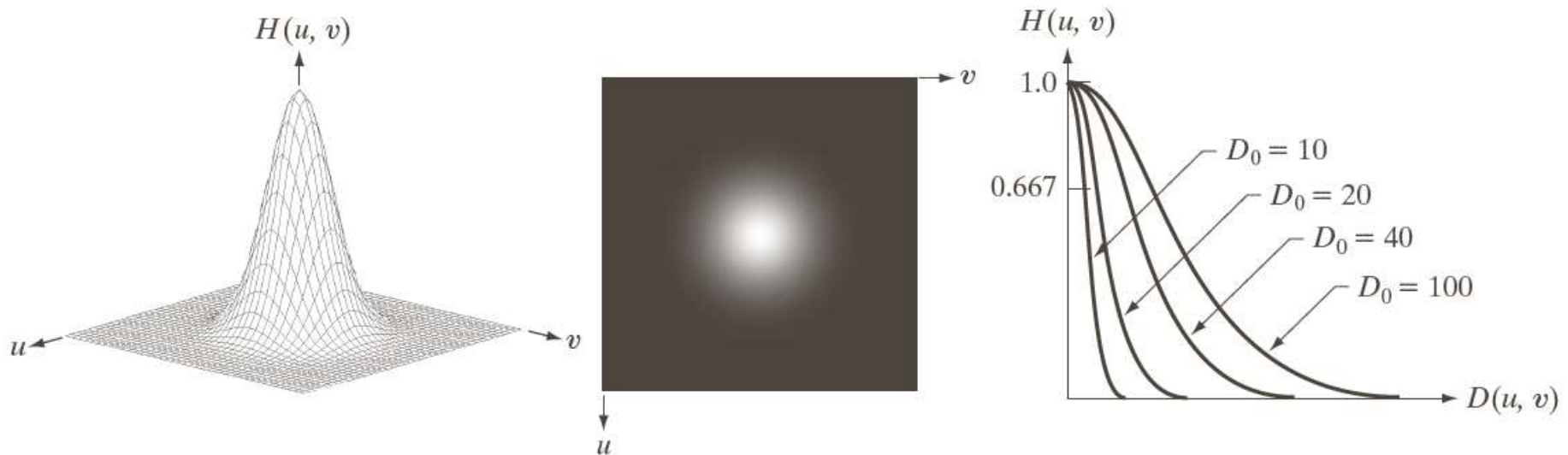
Gaussian Lowpass Filters (GLPF) in two dimensions is given

$$H(u, v) = e^{-D^2(u, v)/2\sigma^2}$$

By letting $\sigma = D_0$

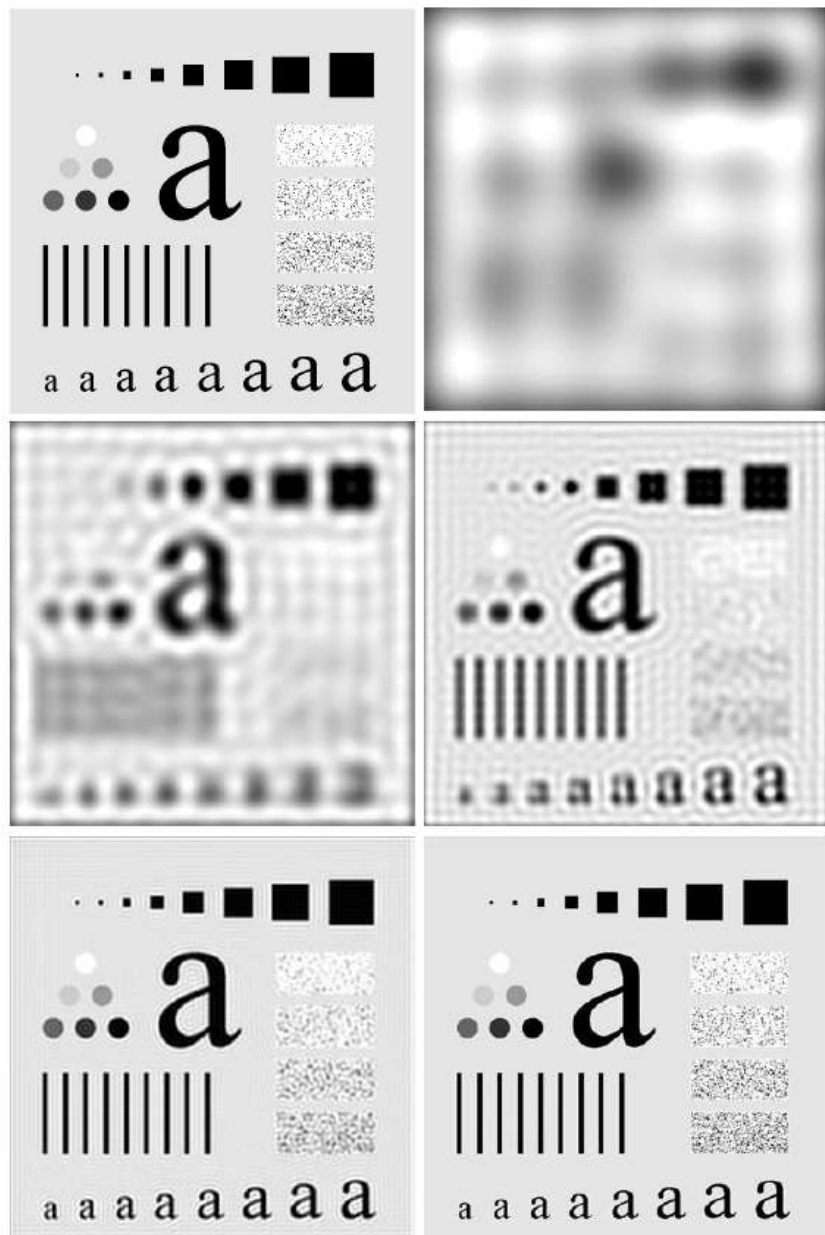
$$H(u, v) = e^{-D^2(u, v)/2D_0^2}$$

Image Smoothing Using Frequency Domain Filters: GLPF



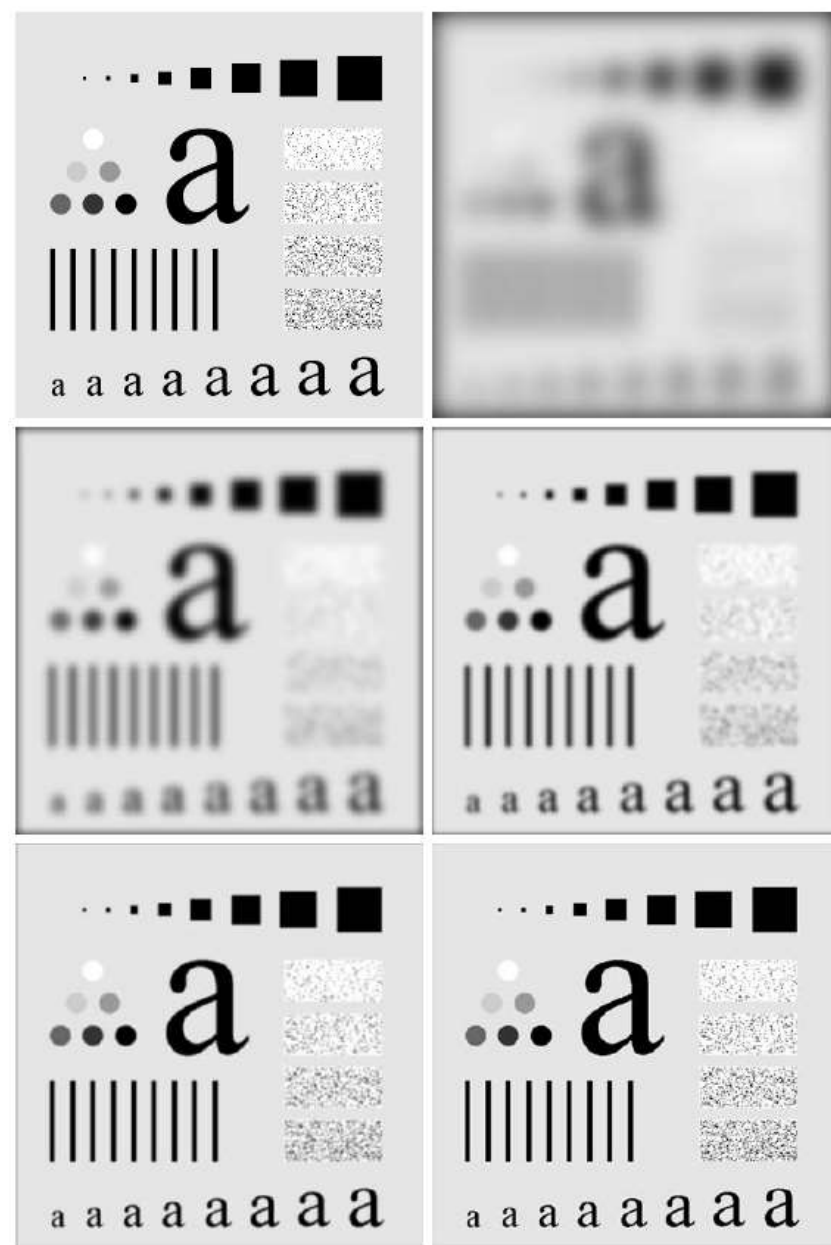
a b c

FIGURE 4.47 (a) Perspective plot of a GLPF transfer function. (b) Filter displayed as an image. (c) Filter radial cross sections for various values of D_0 .



a b
c d
e f

FIGURE 4.42 (a) Original image. (b)–(f) Results of filtering using ILPFs with cutoff frequencies set at radii values 10, 30, 60, 160, and 460, as shown in Fig. 4.41(b). The power removed by these filters was 13, 6.9, 4.3, 2.2, and 0.8% of the total, respectively.



a b
c d
e f

FIGURE 4.48 (a) Original image. (b)–(f) Results of filtering using GLPFs with cutoff frequencies at the radii shown in Fig. 4.41. Compare with Figs. 4.42 and 4.45.

Examples of Smoothing by GLPF (1)

Historically, certain computer programs were written using only two digits rather than four to define the applicable year. Accordingly, the company's software may recognize a date using "00" as 1900 rather than the year 2000.



Historically, certain computer programs were written using only two digits rather than four to define the applicable year. Accordingly, the company's software may recognize a date using "00" as 1900 rather than the year 2000.



a b

FIGURE 4.49

(a) Sample text of low resolution (note broken characters in magnified view). (b) Result of filtering with a GLPF (broken character segments were joined).

Examples of Smoothing by GLPF (2)



FIGURE 4.50 (a) Original image (784×732 pixels). (b) Result of filtering using a GLPF with $D_0 = 100$. (c) Result of filtering using a GLPF with $D_0 = 80$. Note the reduction in fine skin lines in the magnified sections in (b) and (c).

Examples of Smoothing by GLPF (3)



a b c

FIGURE 4.51 (a) Image showing prominent horizontal scan lines. (b) Result of filtering using a GLPF with $D_0 = 50$. (c) Result of using a GLPF with $D_0 = 20$. (Original image courtesy of NOAA.)

Image Sharpening Using Frequency Domain Filters

A highpass filter is obtained from a given lowpass filter using

$$H_{HP}(u, v) = 1 - H_{LP}(u, v)$$

A 2-D ideal highpass filter (IHPL) is defined as

$$H(u, v) = \begin{cases} 0 & \text{if } D(u, v) \leq D_0 \\ 1 & \text{if } D(u, v) > D_0 \end{cases}$$

Image Sharpening Using Frequency Domain Filters

A 2-D Butterworth highpass filter (BHPL) is defined as

$$H(u, v) = \frac{1}{1 + [D_0 / D(u, v)]^{2n}}$$

A 2-D Gaussian highpass filter (GHPL) is defined as

$$H(u, v) = 1 - e^{-D^2(u, v) / 2D_0^2}$$

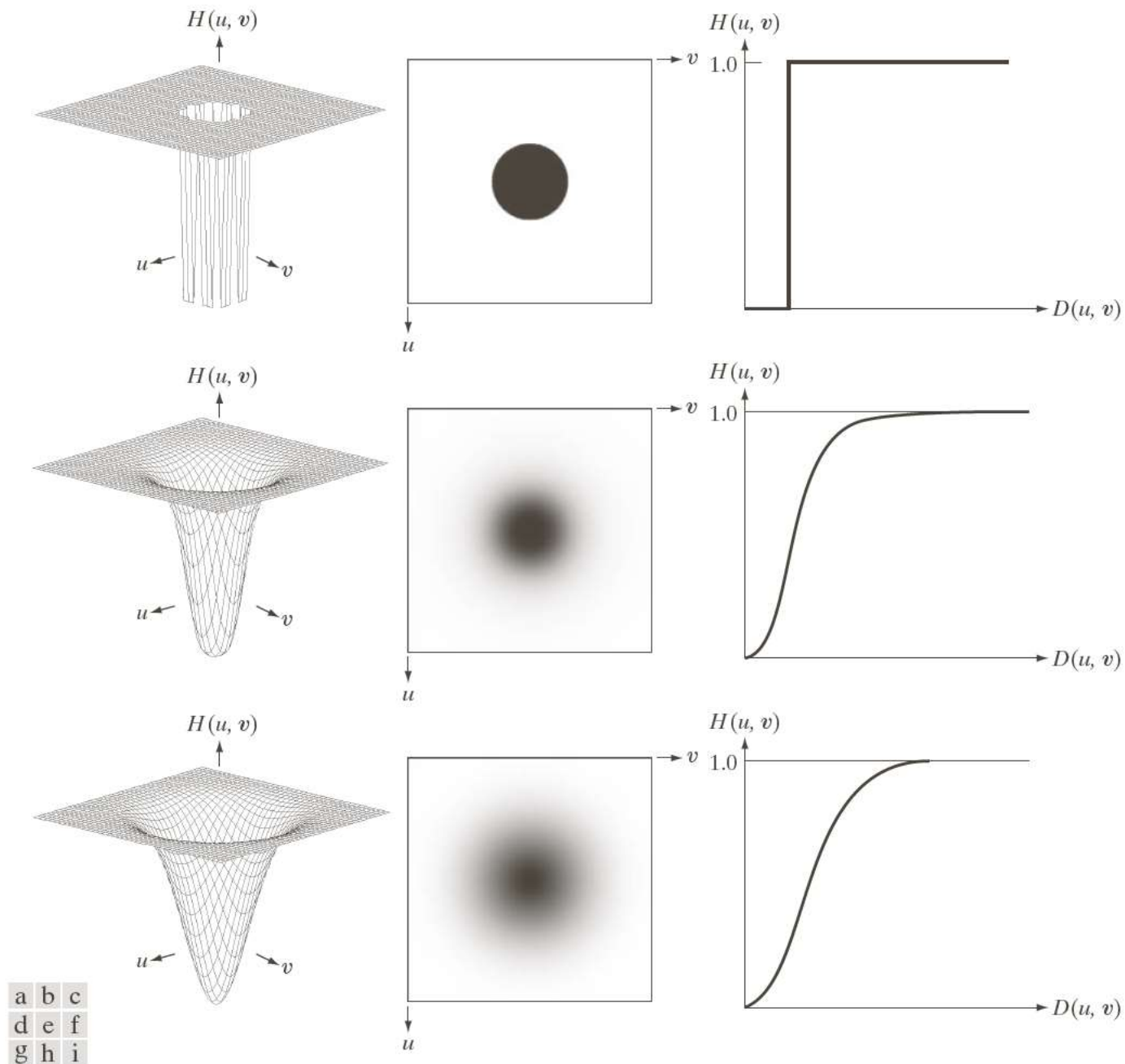


FIGURE 4.52 Top row: Perspective plot, image representation, and cross section of a typical ideal highpass filter. Middle and bottom rows: The same sequence for typical Butterworth and Gaussian highpass filters.

The Spatial Representation of Highpass Filters

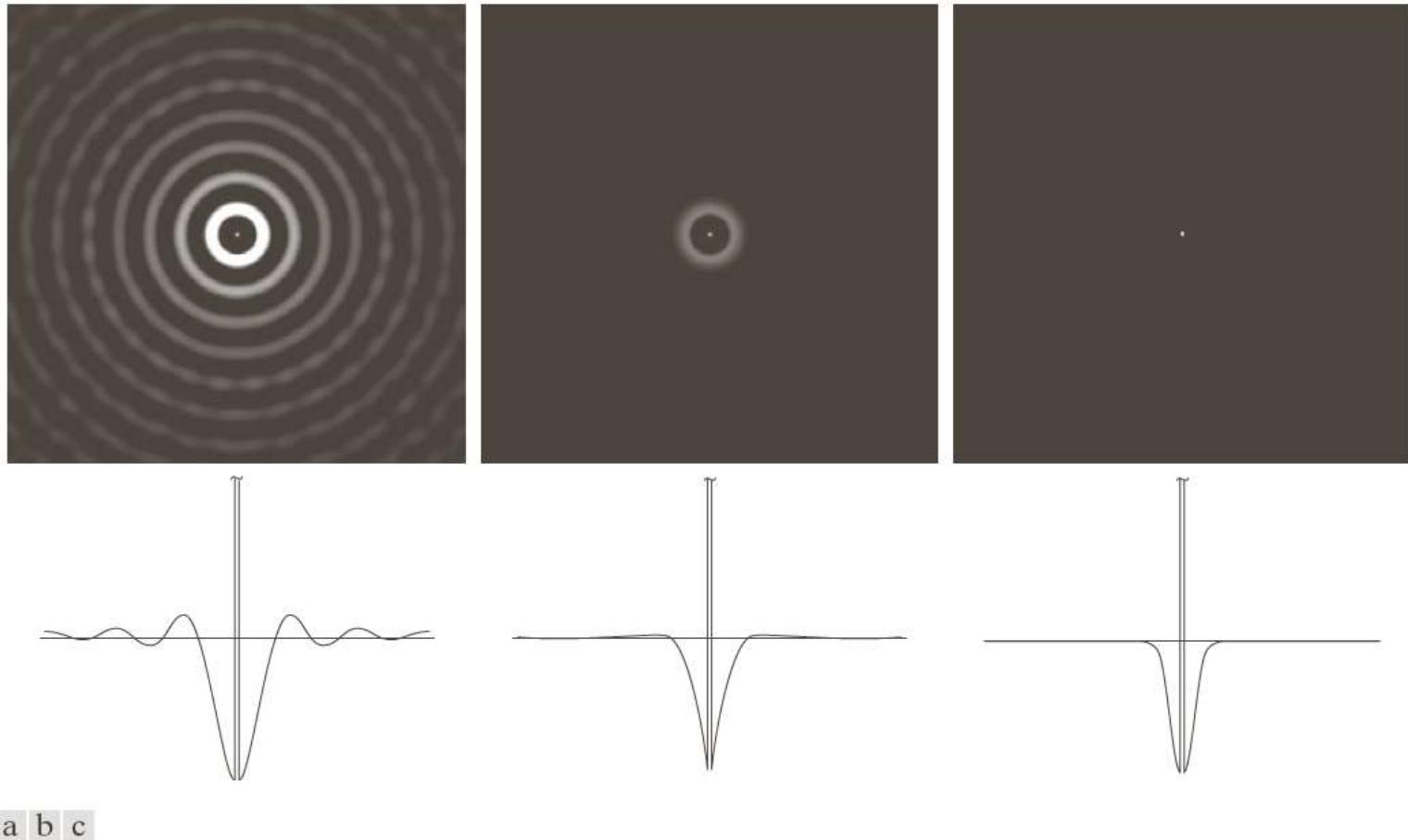


FIGURE 4.53 Spatial representation of typical (a) ideal, (b) Butterworth, and (c) Gaussian frequency domain highpass filters, and corresponding intensity profiles through their centers.

Filtering Results by IHPF



a b c

FIGURE 4.54 Results of highpass filtering the image in Fig. 4.41(a) using an IHPF with $D_0 = 30, 60$, and 160 .

Filtering Results by BHPF



a b c

FIGURE 4.55 Results of highpass filtering the image in Fig. 4.41(a) using a BHPF of order 2 with $D_0 = 30, 60,$ and 160 , corresponding to the circles in Fig. 4.41(b). These results are much smoother than those obtained with an IHPF.

Filtering Results by GHPF



a b c

FIGURE 4.56 Results of highpass filtering the image in Fig. 4.41(a) using a GHPF with $D_0 = 30, 60,$ and $160,$ corresponding to the circles in Fig. 4.41(b). Compare with Figs. 4.54 and 4.55.

Using Highpass Filtering and Threshold for Image Enhancement

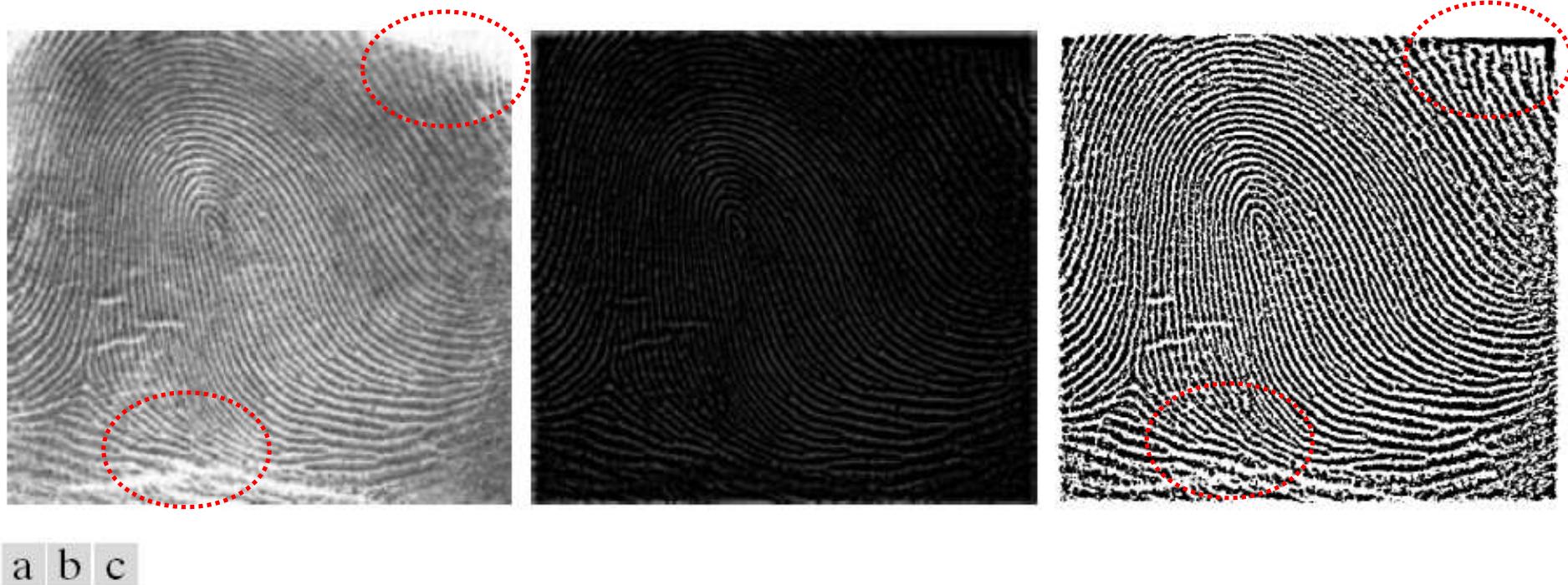


FIGURE 4.57 (a) Thumb print. (b) Result of highpass filtering (a). (c) Result of thresholding (b). (Original image courtesy of the U.S. National Institute of Standards and Technology.)

The Laplacian in the Frequency Domain

$$H(u, v) = -4\pi^2(u^2 + v^2)$$

$$\begin{aligned} H(u, v) &= -4\pi^2 \left[(u - P/2)^2 + (v - Q/2)^2 \right] \\ &= -4\pi^2 D^2(u, v) \end{aligned}$$

The Laplacian image

$$\nabla^2 f(x, y) = \mathfrak{F}^{-1} \{ H(u, v) F(u, v) \}$$

Enhancement is obtained

$$g(x, y) = f(x, y) + c \nabla^2 f(x, y) \quad c = -1$$

The Laplacian in the Frequency Domain

The enhanced image

$$\begin{aligned} g(x, y) &= \mathfrak{F}^{-1} \{ F(u, v) - H(u, v)F(u, v) \} \\ &= \mathfrak{F}^{-1} \{ [1 - H(u, v)] F(u, v) \} \\ &= \mathfrak{F}^{-1} \{ [1 + 4\pi^2 D^2(u, v)] F(u, v) \} \end{aligned}$$

The Laplacian in the Frequency Domain



a b

FIGURE 4.58

(a) Original, blurry image.

(b) Image enhanced using the Laplacian in the frequency domain. Compare with Fig. 3.38(e).

Unsharp Masking, Highboost Filtering and High-Frequency-Emphasis Filtering

$$g_{mask}(x, y) = f(x, y) - f_{LP}(x, y)$$

$$f_{LP}(x, y) = \mathfrak{F}^{-1} [H_{LP}(u, v)F(u, v)]$$

Unsharp masking and highboost filtering

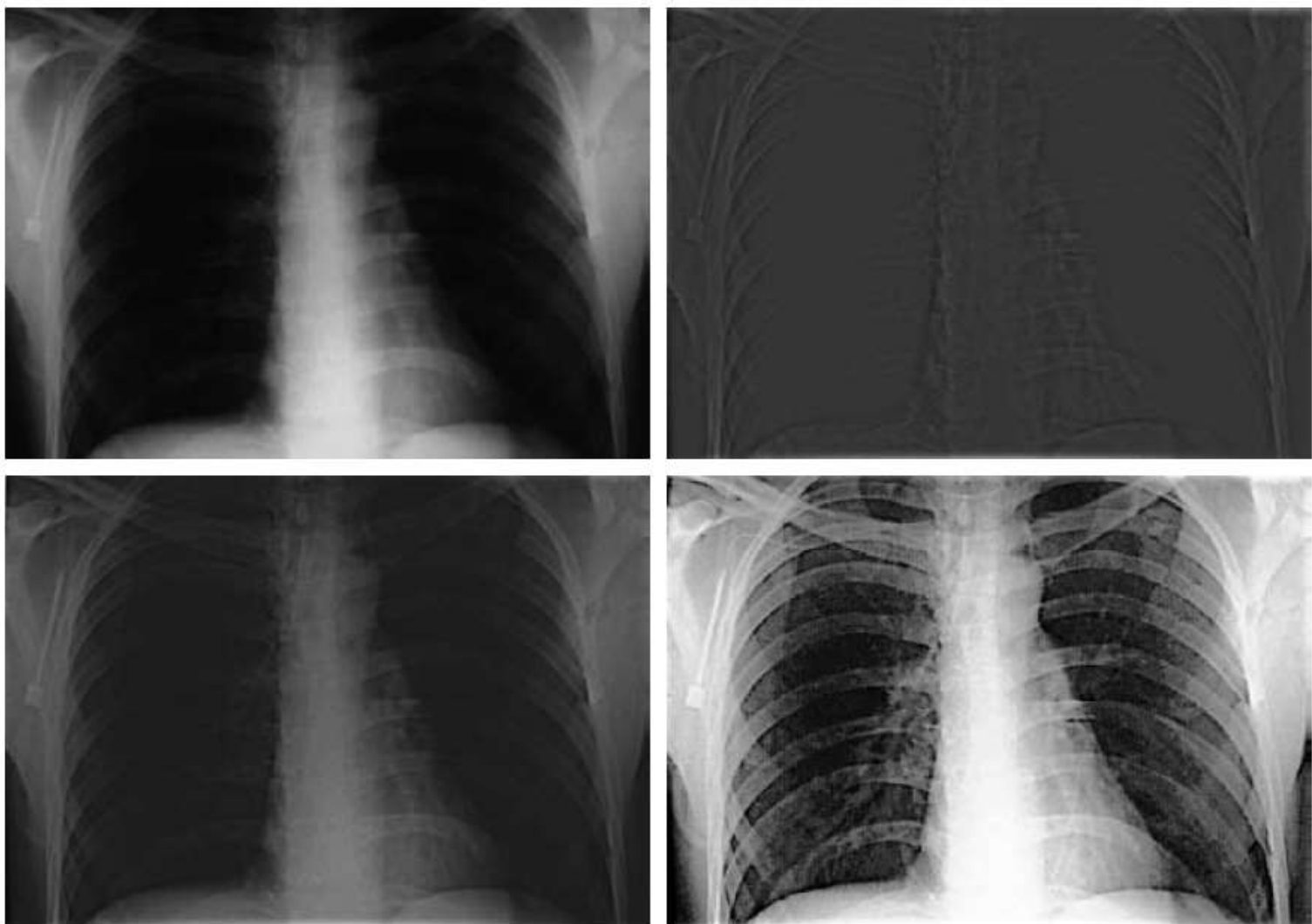
$$g(x, y) = f(x, y) + k * g_{mask}(x, y)$$

$$\begin{aligned} g(x, y) &= \mathfrak{F}^{-1} \left\{ [1 + k * [1 - H_{LP}(u, v)]] F(u, v) \right\} \\ &= \mathfrak{F}^{-1} \left\{ [1 + k * H_{HP}(u, v)] F(u, v) \right\} \end{aligned}$$

Unsharp Masking, Highboost Filtering and High-Frequency-Emphasis Filtering

$$g(x, y) = \mathfrak{F}^{-1} \left\{ [k_1 + k_2 * H_{HP}(u, v)] F(u, v) \right\}$$

$$k_1 \geq 0 \quad \text{and} \quad k_2 \geq 0$$



a	b
c	d

FIGURE 4.59 (a) A chest X-ray image. (b) Result of highpass filtering with a Gaussian filter. (c) Result of high-frequency-emphasis filtering using the same filter. (d) Result of performing histogram equalization on (c). (Original image courtesy of Dr. Thomas R. Gest, Division of Anatomical Sciences, University of Michigan Medical School.)

Homomorphic Filtering

$$\begin{aligned} f(x, y) &= i(x, y)r(x, y) \\ &= \mathfrak{I}[f(x, y)] \neq \mathfrak{I}[i(x, y)]\mathfrak{I}[r(x, y)] \end{aligned}$$

$$z(x, y) = \ln f(x, y) = \ln i(x, y) + \ln r(x, y)$$

$$\mathfrak{I}\{z(x, y)\} = \mathfrak{I}\{\ln f(x, y)\} = \mathfrak{I}\{\ln i(x, y)\} + \mathfrak{I}\{\ln r(x, y)\}$$

$$Z(u, v) = F_i(u, v) + F_r(u, v)$$

Homomorphic Filtering

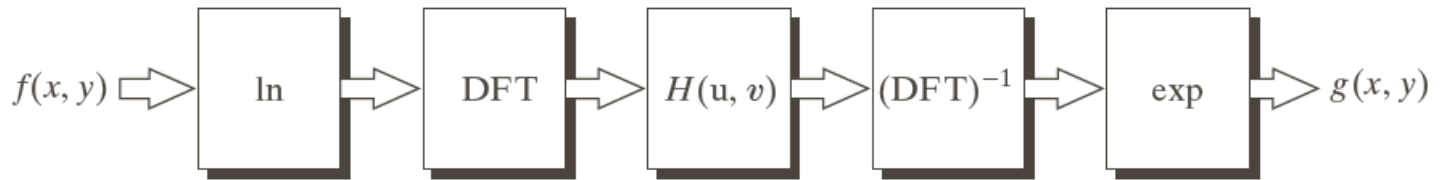
$$\begin{aligned} S(u, v) &= H(u, v)Z(u, v) \\ &= H(u, v)F_i(u, v) + H(u, v)F_r(u, v) \end{aligned}$$

$$\begin{aligned} s(x, y) &= \mathfrak{F}^{-1}\{S(u, v)\} \\ &= \mathfrak{F}^{-1}\{H(u, v)F_i(u, v) + H(u, v)F_r(u, v)\} \\ &= \mathfrak{F}^{-1}\{H(u, v)F_i(u, v)\} + \mathfrak{F}^{-1}\{H(u, v)F_r(u, v)\} \\ &= i'(x, y) + r'(x, y) \end{aligned}$$

$$g(x, y) = e^{s(x, y)} = e^{i'(x, y)} e^{r'(x, y)} = i_0(x, y) r_0(x, y)$$

Homomorphic Filtering

FIGURE 4.60
Summary of steps
in homomorphic
filtering.



The illumination component of an image generally is characterized by slow spatial variations, while the reflectance component tends to vary abruptly

These characteristics lead to associating the low frequencies of the Fourier transform of the logarithm of an image with illumination the high frequencies with reflectance.

Homomorphic Filtering

$$H(u, v) = (\gamma_H - \gamma_L) \left[1 - e^{-c \left[D^2(u, v) / D_0^2 \right]} \right] + \gamma_L$$

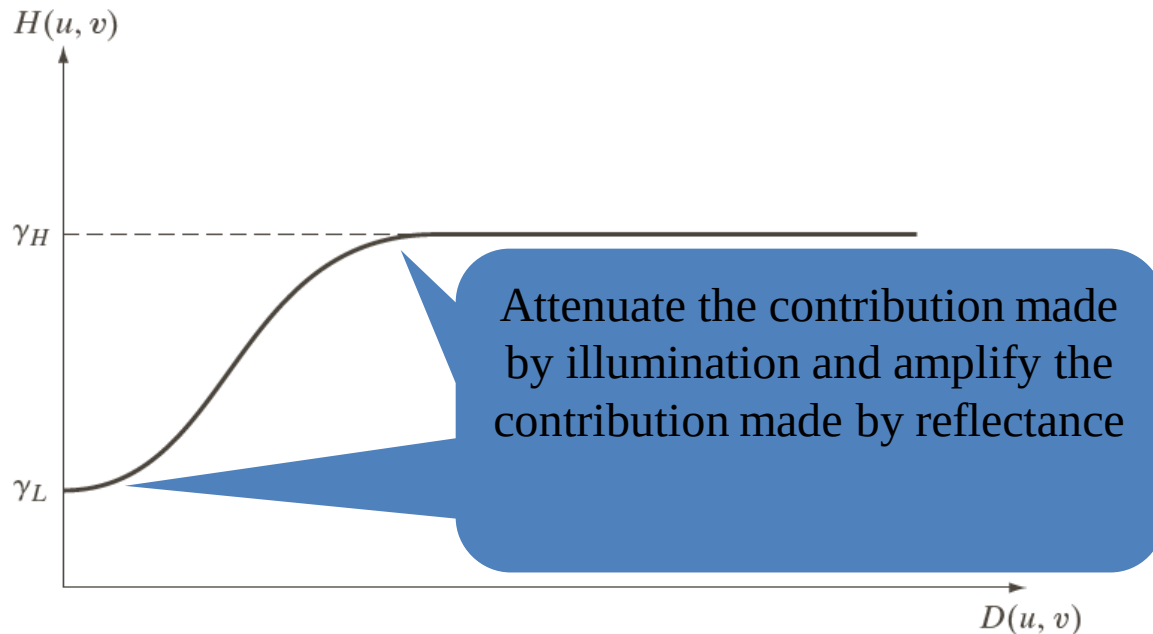


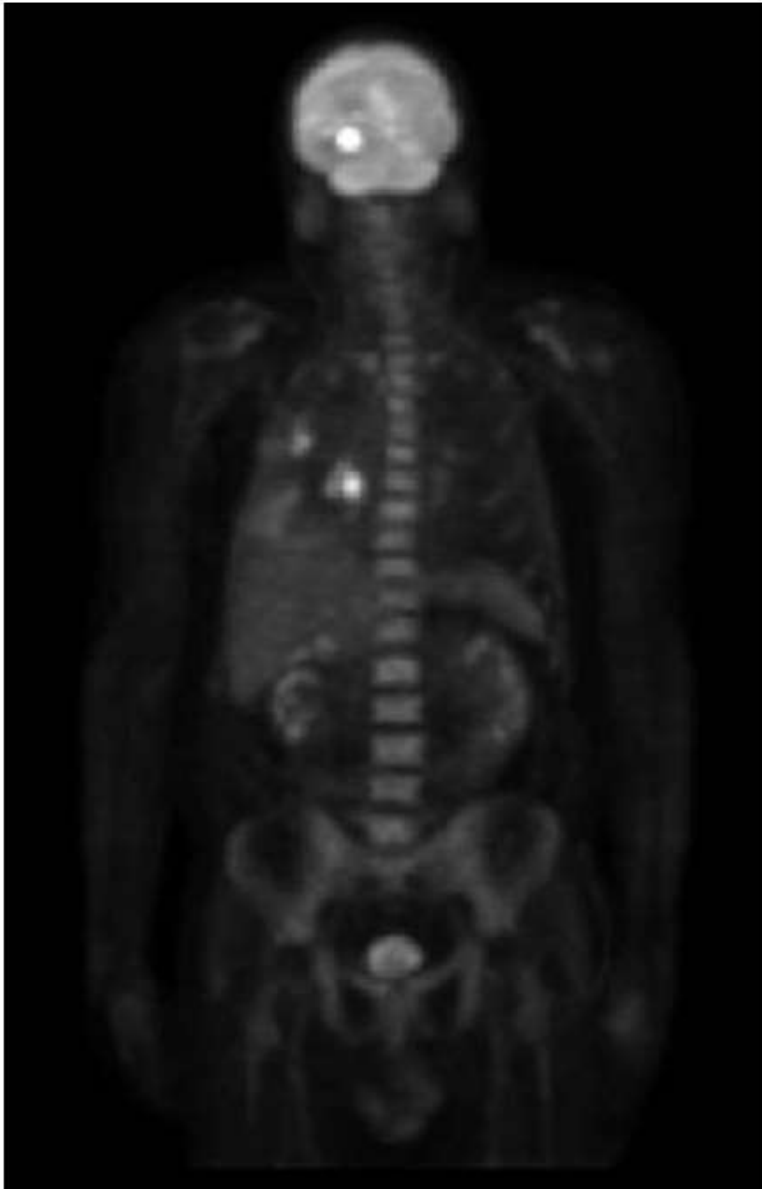
FIGURE 4.61 Radial cross section of a circularly symmetric homomorphic filter function. The vertical axis is at the center of the frequency rectangle and $D(u, v)$ is the distance from the center.

Homomorphic Filtering

a b

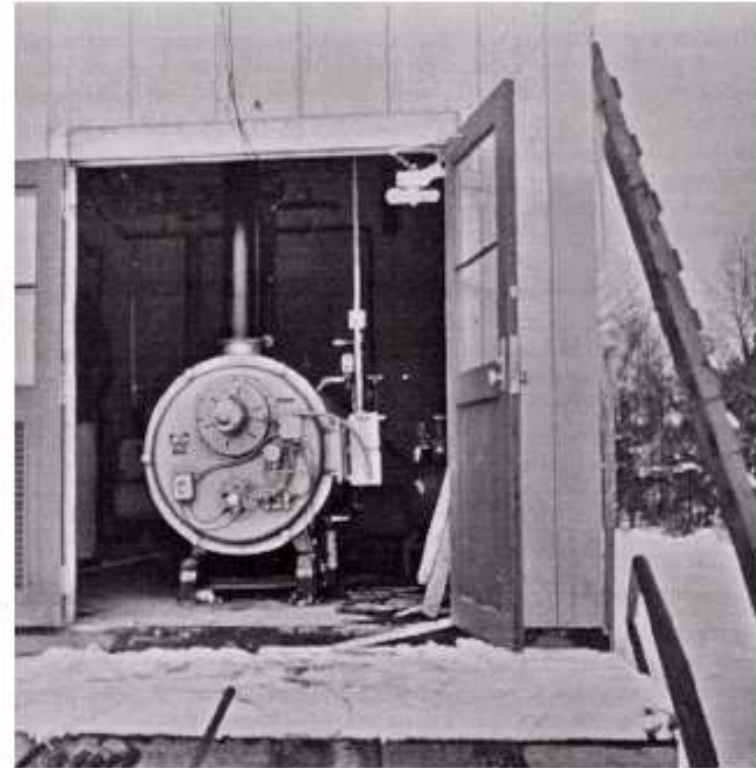
FIGURE 4.62

(a) Full body PET scan. (b) Image enhanced using homomorphic filtering. (Original image courtesy of Dr. Michael E. Casey, CTI PET Systems.)



Homomorphic Filtering

a b
FIGURE
(a) Original image. (b) Image processed by homomorphic filtering (note details inside shelter). (Stockham.)



Selective Filtering

Non-Selective Filters:

operate over the entire frequency rectangle

Selective Filters

operate over some part, not entire frequency rectangle

- **bandreject or bandpass:** process specific bands
- **notch filters:** process small regions of the frequency rectangle

Selective Filtering: Bandreject and Bandpass Filters

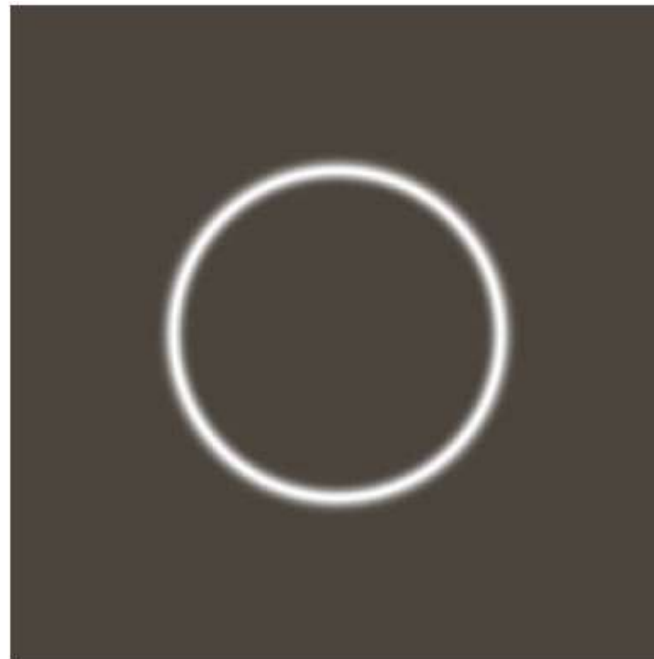
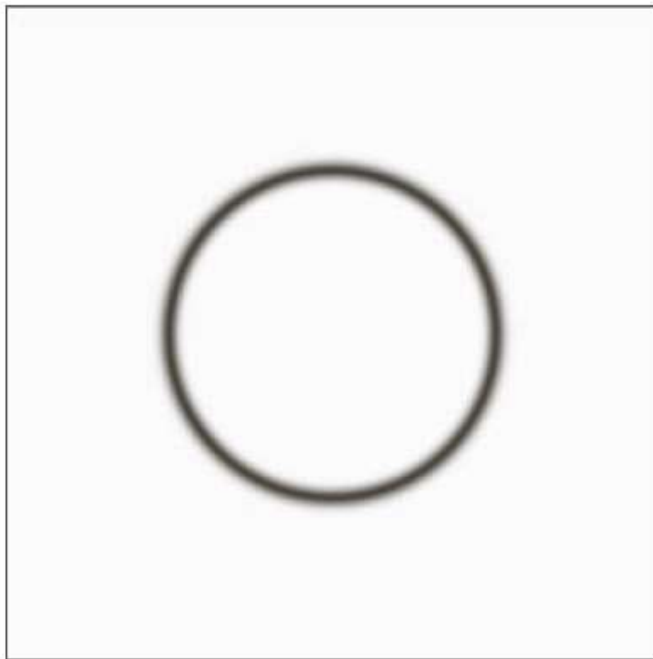
TABLE 4.6

Bandreject filters. W is the width of the band, D is the distance $D(u, v)$ from the center of the filter, D_0 is the cutoff frequency, and n is the order of the Butterworth filter. We show D instead of $D(u, v)$ to simplify the notation in the table.

Ideal	Butterworth	Gaussian
$H(u, v) = \begin{cases} 0 & \text{if } D_0 - \frac{W}{2} \leq D \leq D_0 + \frac{W}{2} \\ 1 & \text{otherwise} \end{cases}$	$H(u, v) = \frac{1}{1 + \left[\frac{DW}{D^2 - D_0^2} \right]^{2n}}$	$H(u, v) = 1 - e^{-\left[\frac{D^2 - D_0^2}{DW} \right]^2}$

$$H_{BP}(u, v) = 1 - H_{BR}(u, v)$$

Selective Filtering: Bandreject and Bandpass Filters



a b

FIGURE 4.63

(a) Bandreject Gaussian filter.
(b) Corresponding bandpass filter.
The thin black border in (a) was added for clarity; it is not part of the data.

Selective Filtering: Notch Filters

Zero-phase-shift filters must be symmetric about the origin.

A notch with center at (u_0, v_0) must have a corresponding notch at location $(-u_0, -v_0)$.

Notch reject filters are constructed as products of highpass filters whose centers have been translated to the centers of the notches.

$$H_{NR}(u, v) = \prod_{k=1}^Q H_k(u, v) H_{-k}(u, v)$$

where $H_k(u, v)$ and $H_{-k}(u, v)$ are highpass filters whose centers are at (u_k, v_k) and $(-u_k, -v_k)$, respectively.

Selective Filtering: Notch Filters

$$H_{NR}(u, v) = \prod_{k=1}^Q H_k(u, v) H_{-k}(u, v)$$

where $H_k(u, v)$ and $H_{-k}(u, v)$ are highpass filters whose centers are at (u_k, v_k) and $(-u_k, -v_k)$, respectively.

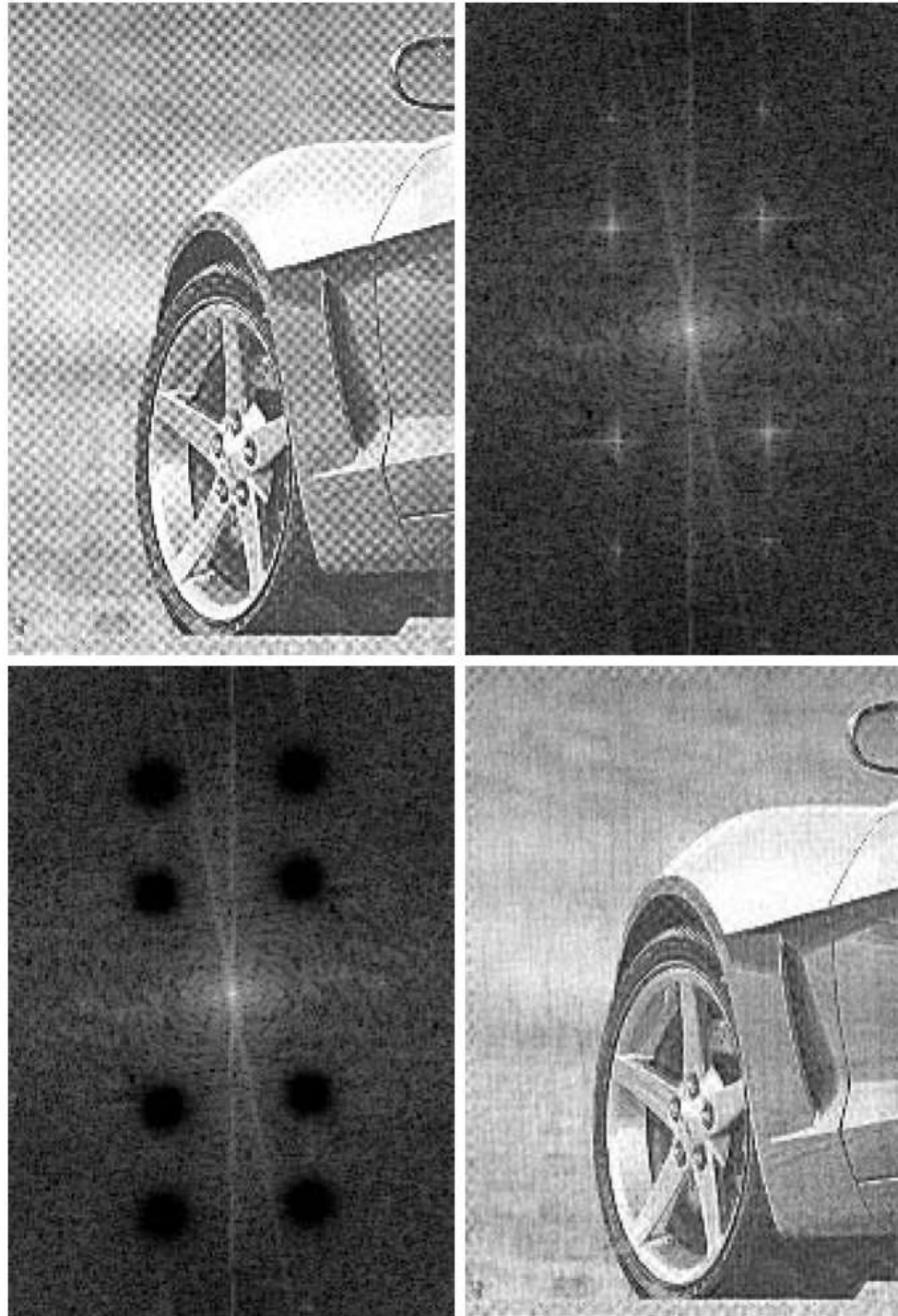
A Butterworth notch reject filter of order n

$$H_{NR}(u, v) = \prod_{k=1}^3 \left[\frac{1}{1 + [D_{0k} / D_k(u, v)]^{2n}} \right] \left[\frac{1}{1 + [D_{0k} / D_{-k}(u, v)]^{2n}} \right]$$

$$D_k(u, v) = \left[(u - M / 2 - u_k)^2 + (v - N / 2 - v_k)^2 \right]^{1/2}$$

$$D_{-k}(u, v) = \left[(u - M / 2 + u_k)^2 + (v - N / 2 + v_k)^2 \right]^{1/2}$$

Examples: Notch Filters (1)



a	b
c	d

FIGURE 4.64

(a) Sampled newspaper image showing a moiré pattern.

(b) Spectrum.

(c) Butterworth notch reject filter multiplied by the Fourier transform.

(d) Filtered image.

A Butterworth notch reject filter $D_0=3$ and $n=4$ for all notch pairs

Examples: Notch Filters (2)

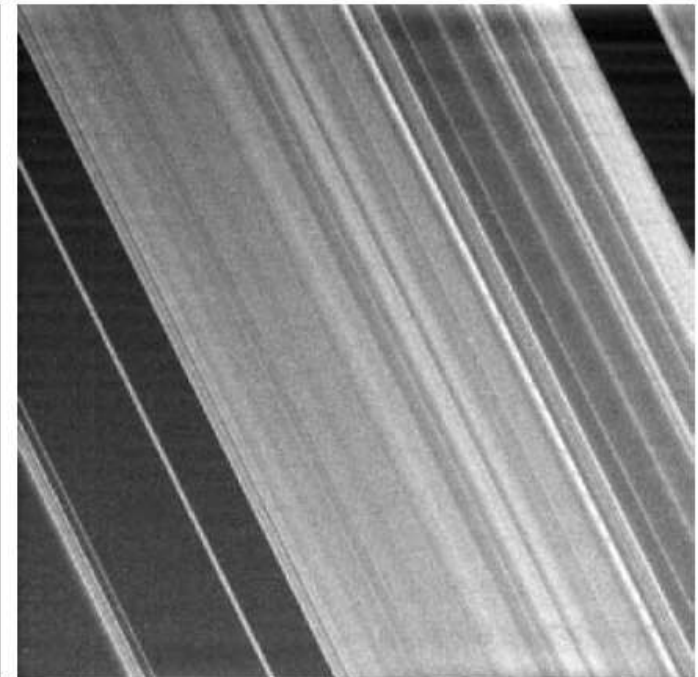
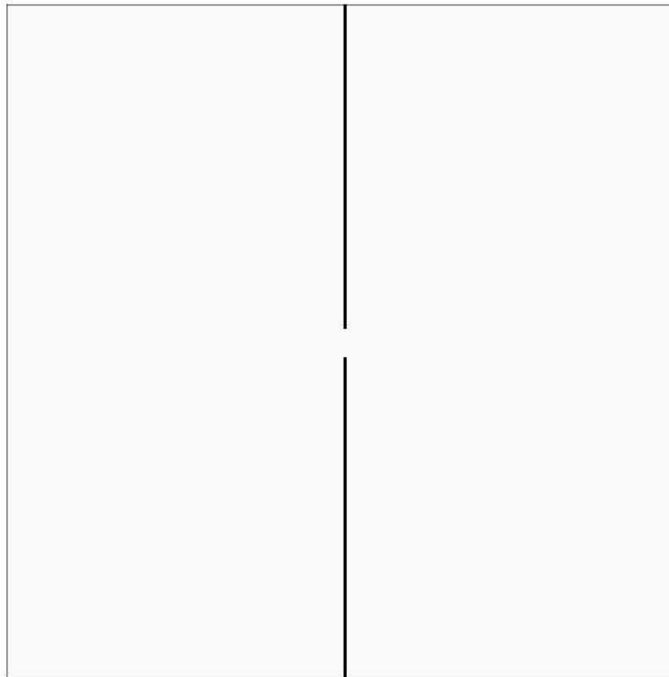
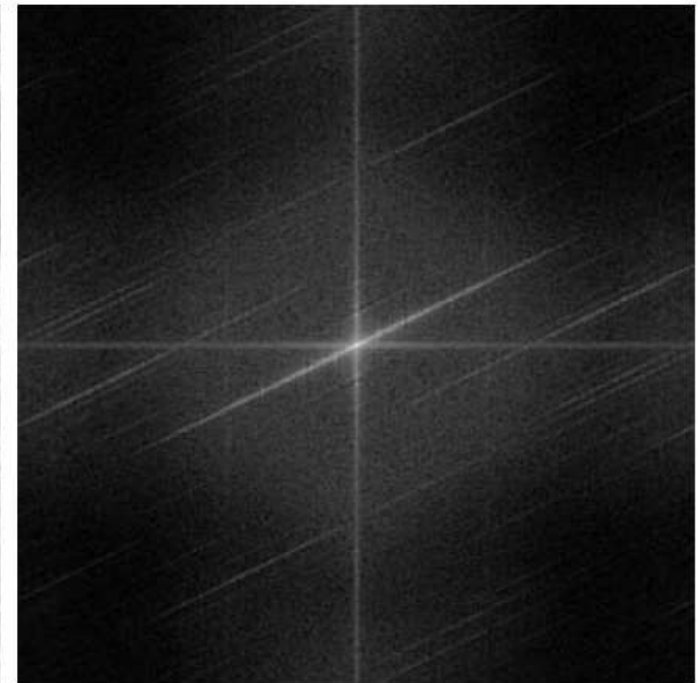
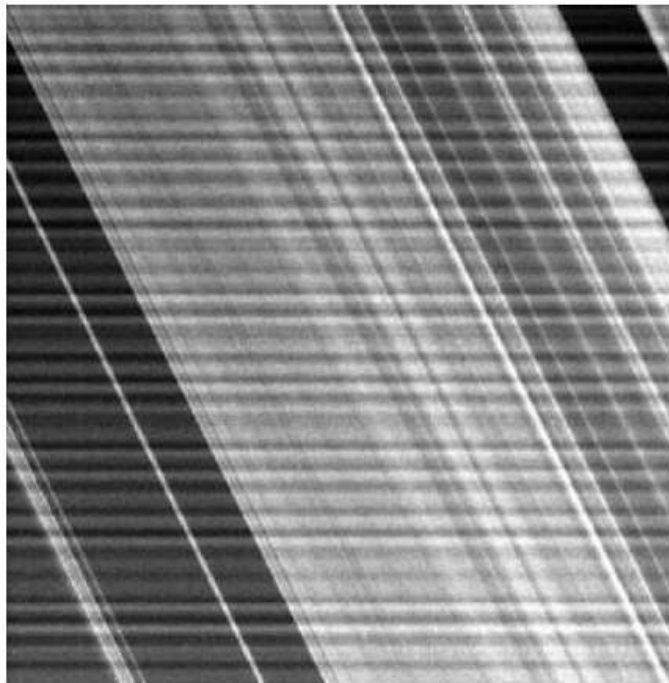
a	b
c	d

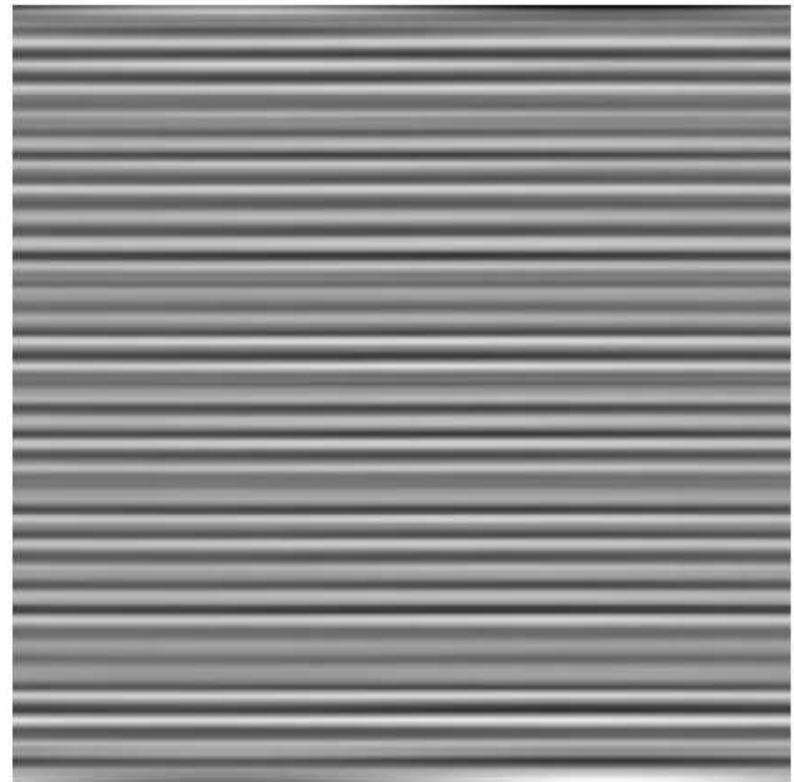
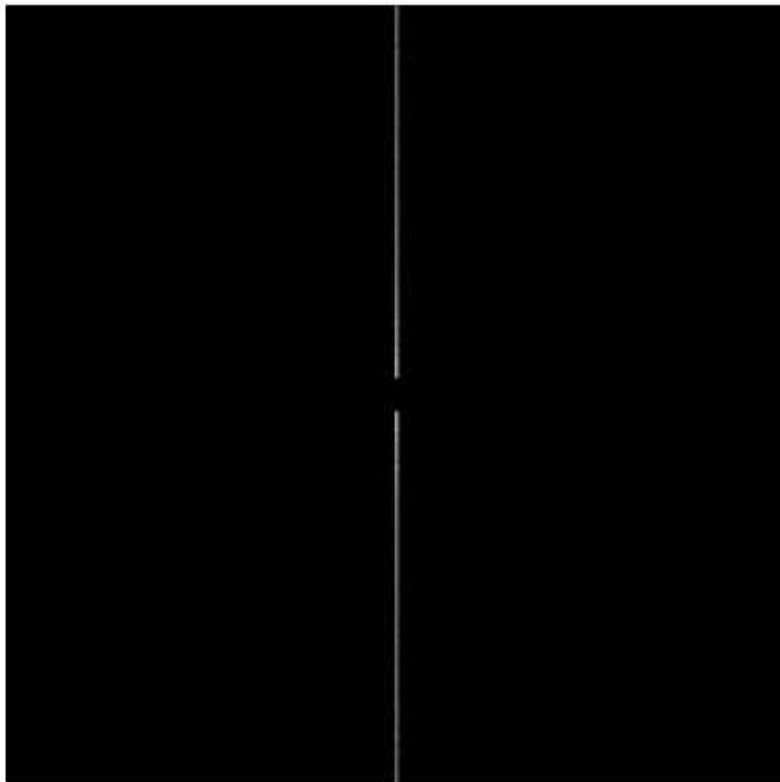
FIGURE 4.65

(a) 674×674 image of the Saturn rings showing nearly periodic interference.

(b) Spectrum: The bursts of energy in the vertical axis near the origin correspond to the interference pattern. (c) A vertical notch reject filter.

(d) Result of filtering. The thin black border in (c) was added for clarity; it is not part of the data. (Original image courtesy of Dr. Robert A. West, NASA/JPL.)





a b

FIGURE 4.66

(a) Result (spectrum) of applying a notch pass filter to the DFT of Fig. 4.65(a).
(b) Spatial pattern obtained by computing the IDFT of (a).

Questions???

Thank You!!!